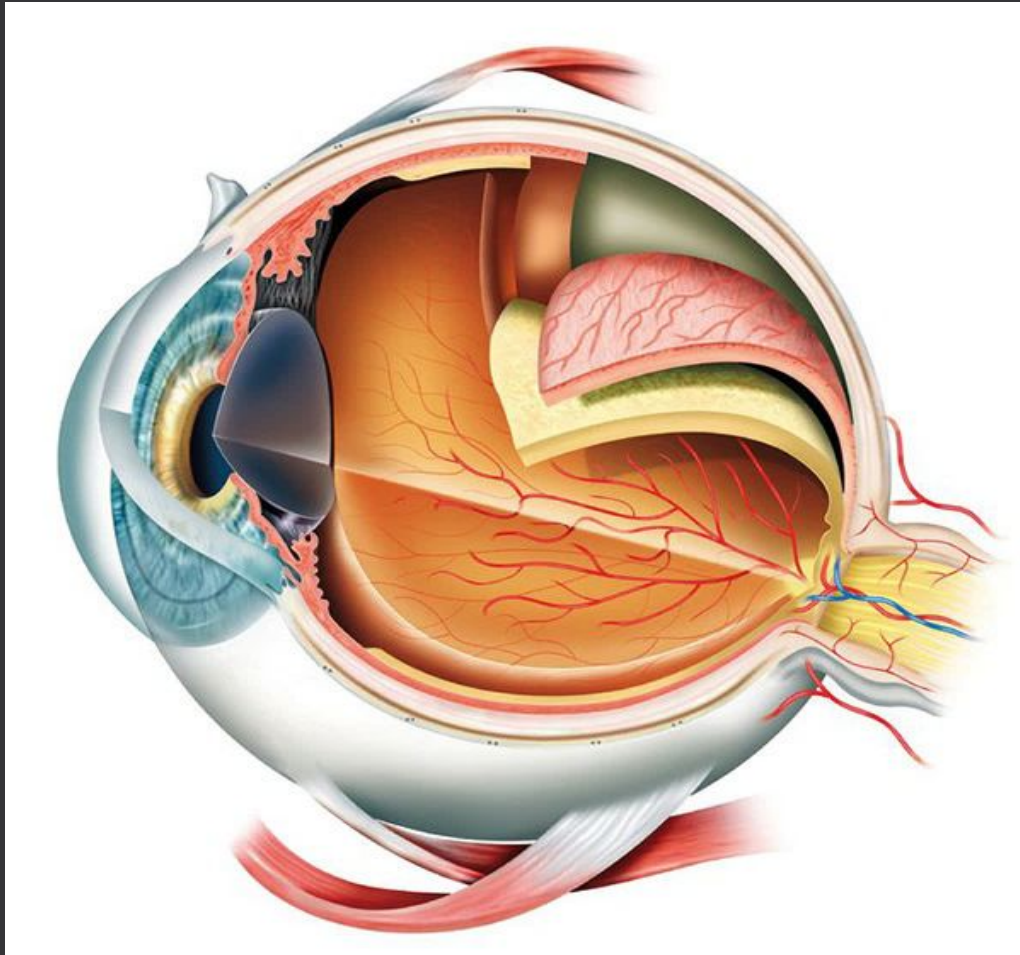


Anatomy & Physiology of The Eye



Piangporn Saksiriwutto, MD.
Department of ophthalmology
Siriraj hospital

Anatomy & Physiology of the eye

Orbit

- Orbital walls
- Orbital apertures
- Extraocular muscles

Ocular adnexa

- Eyelid
- Conjunctiva and tenon's capsule

Lacrimal system

• Eyeball

- Cornea
 - Sclera
 - Lens
 - Aqueous outflow apparatus
 - Uvea
 - Vitreous
 - Retina
- ## • Optic nerve

Orbit - เบ้าตา



Volume ~ 30 cm³

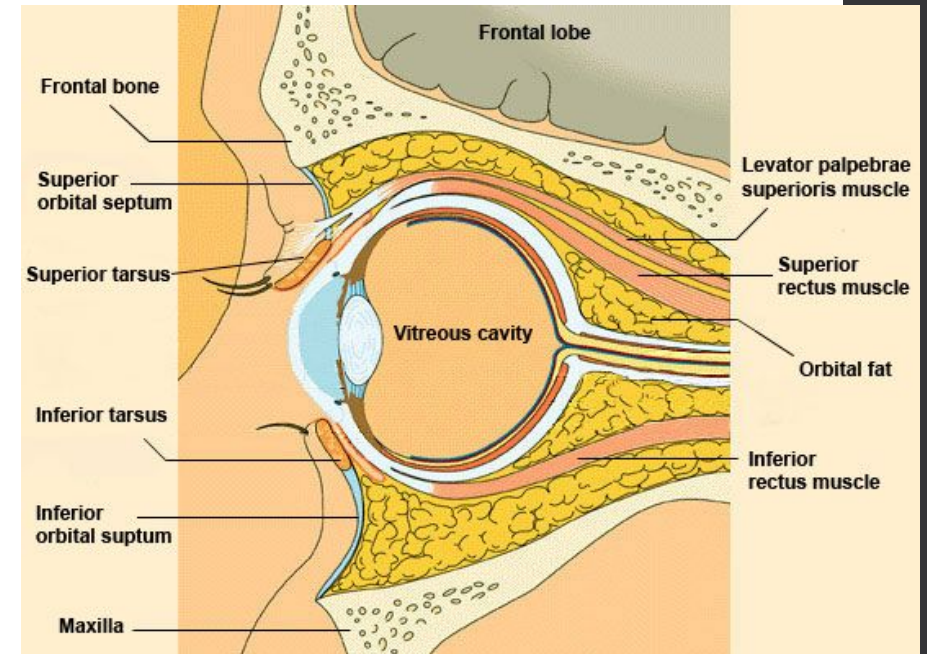
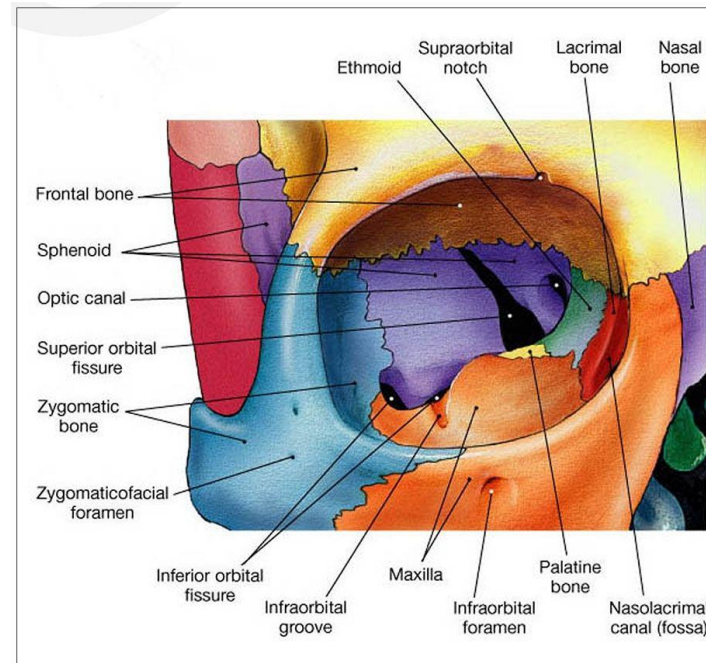
Height 35 mm

Width 45 mm

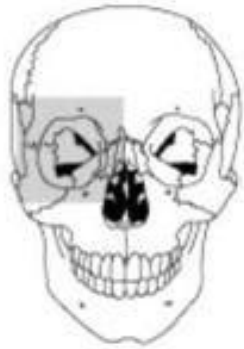
Depth 40-45 mm

•Orbit

- Orbital walls
- Orbital apertures
- Extraocular muscles
- Orbital vessels
- Orbital nerves



Orbital walls



1

Roof of orbit

Lesser wing of sphenoid bone

Orbital plate of frontal bone

3

Lateral wall of orbit

Zygomatic process of frontal bone

Greater wing of sphenoid bone

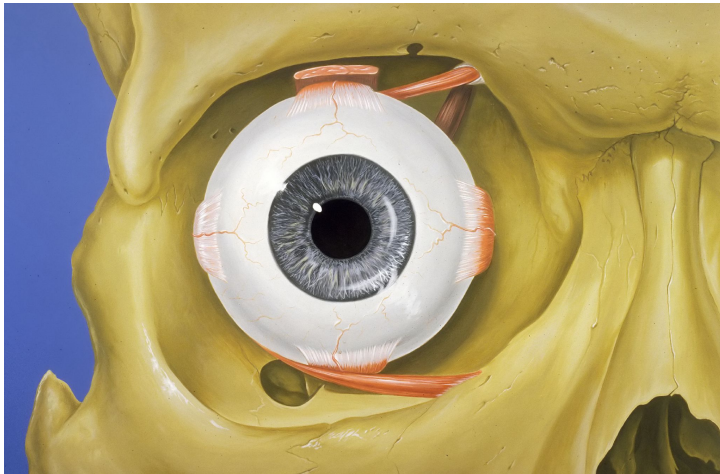
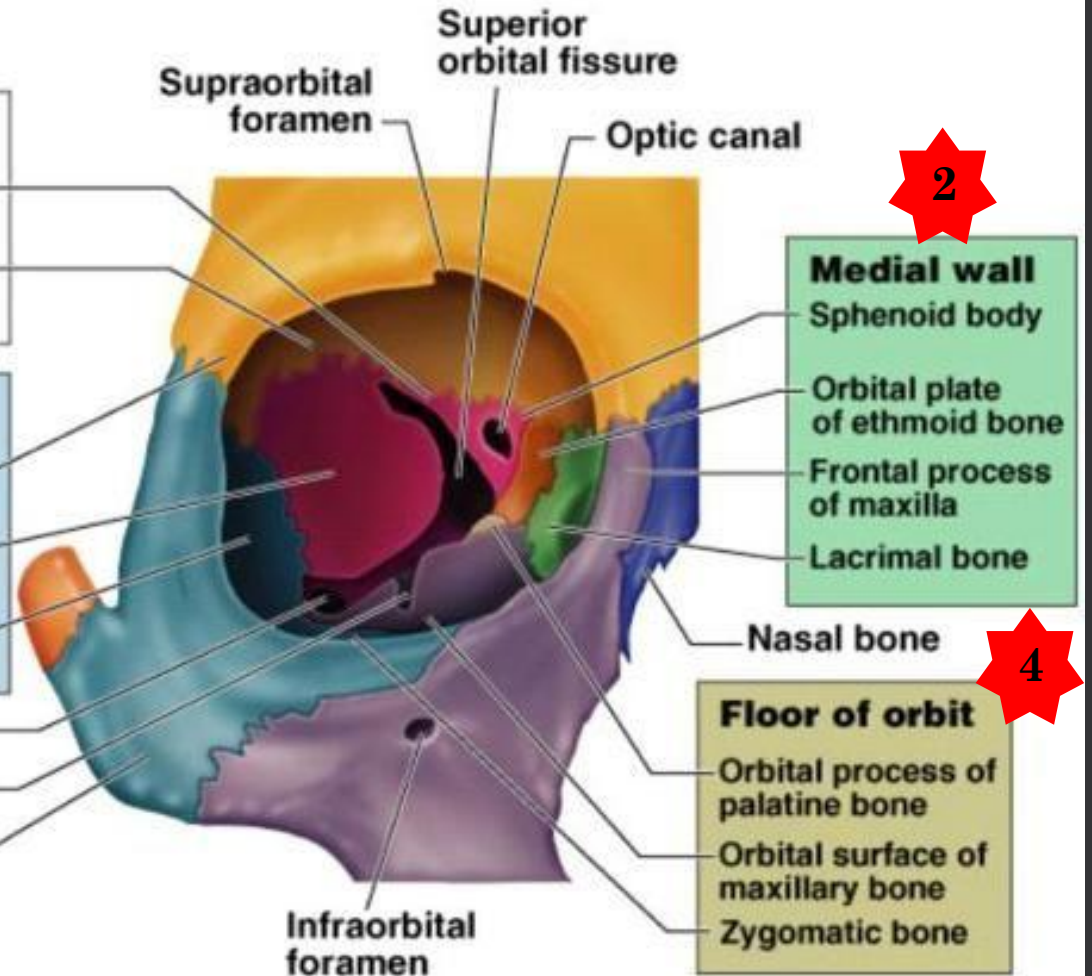
Orbital surface of zygomatic bone

Inferior orbital fissure

Infraorbital groove

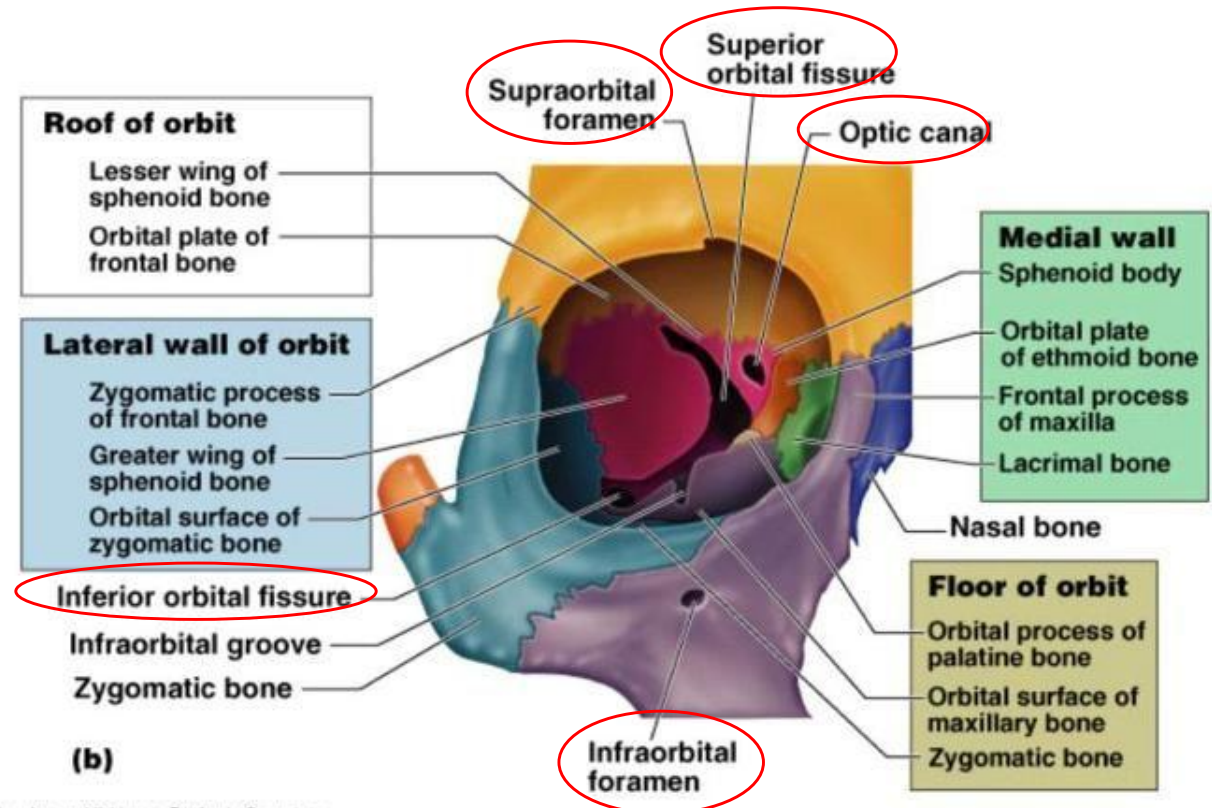
Zygomatic bone

(b)



Orbital apertures

1. Optic canal (foramen)
2. Superior orbital fissure
3. Inferior orbital fissure
4. Supraorbital foramen
5. Infraorbital foramen



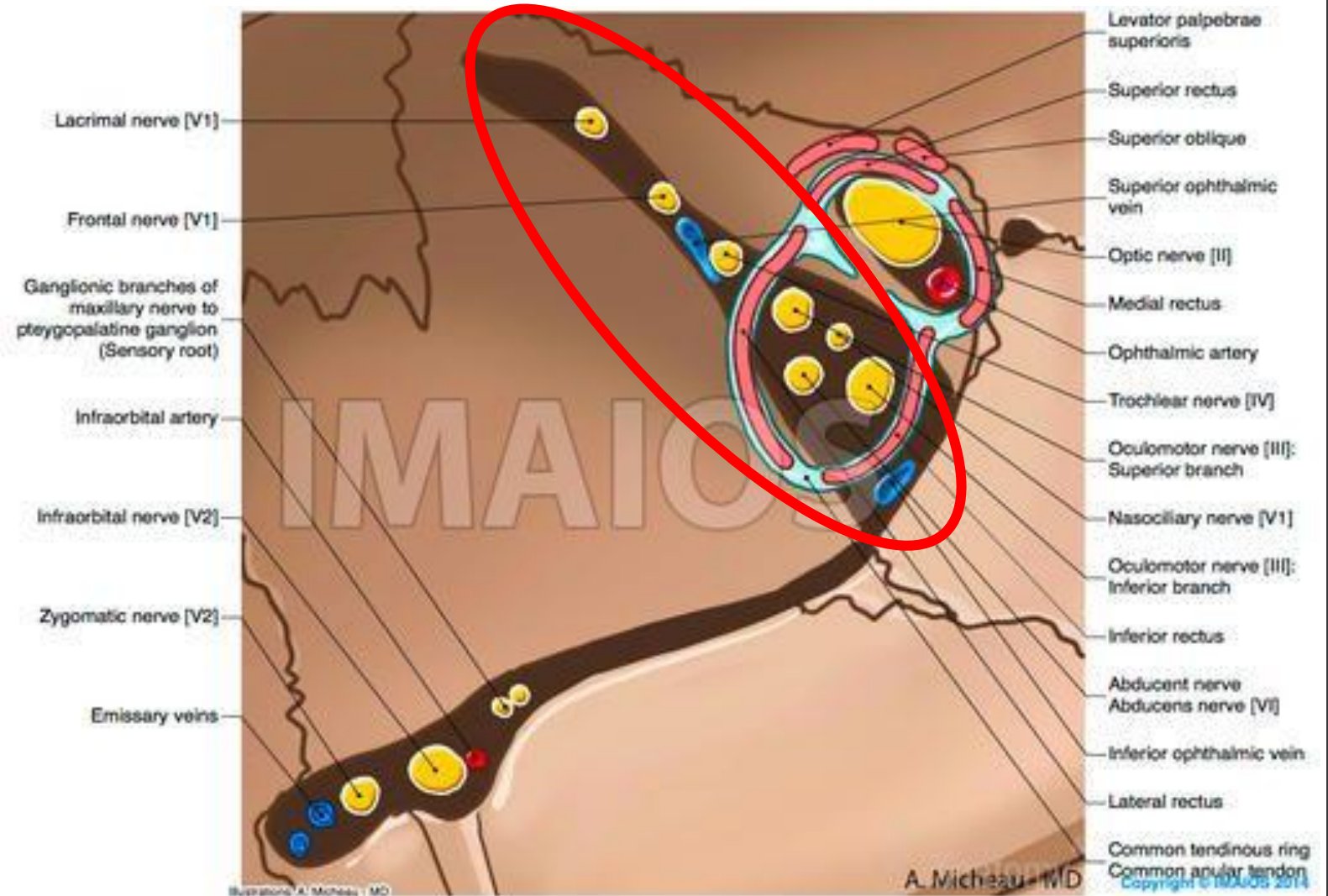
Superior and inferior orbital fissures

Superior orbital fissure

- CN V1 (lacrimal, frontal and nasociliary n.)
- CN III, IV and VI
- Superior ophthalmic vein

Inferior orbital fissure

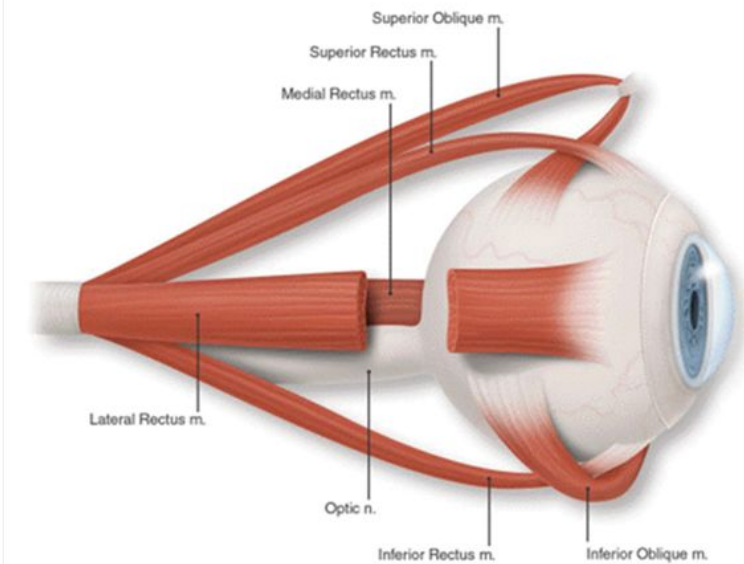
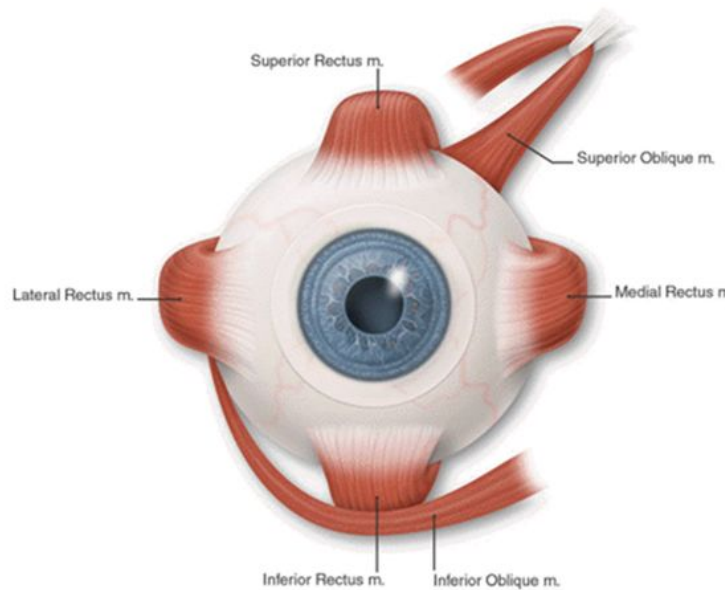
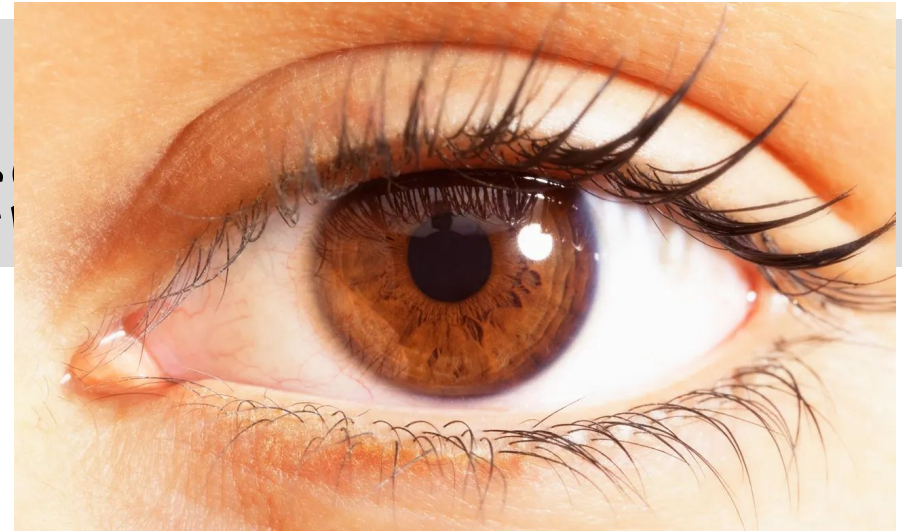
- Infraorbital and zygomatic branches of CN V2
- Orbital n. from pterygopalatine ganglion
- Inferior ophthalmic vein



Extraocular muscles

6 Extraocular muscles

- 4 Rectus muscles
 - Medial rectus
 - Lateral rectus
 - Superior rectus
 - Inferior rectus
- 2 Oblique muscles
 - Superior oblique
 - Inferior oblique



Superior rectus

Superior oblique

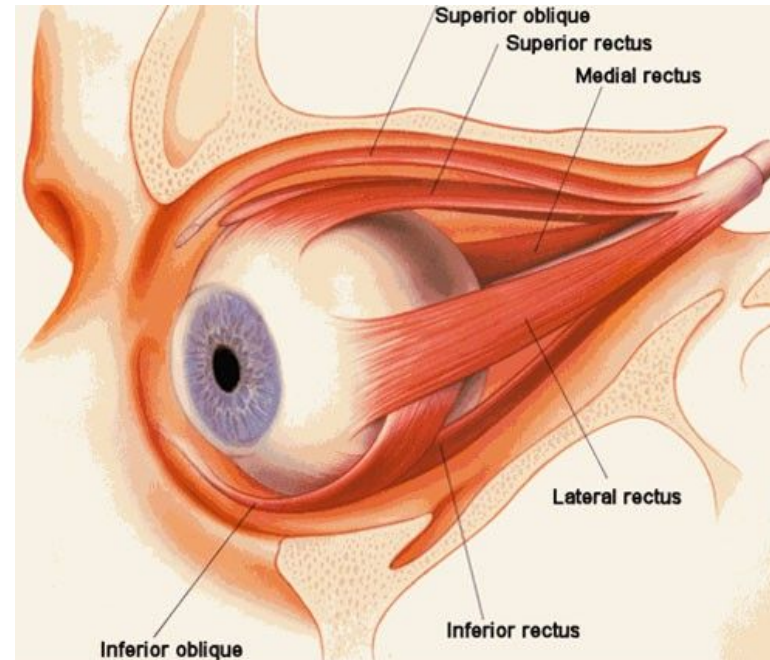
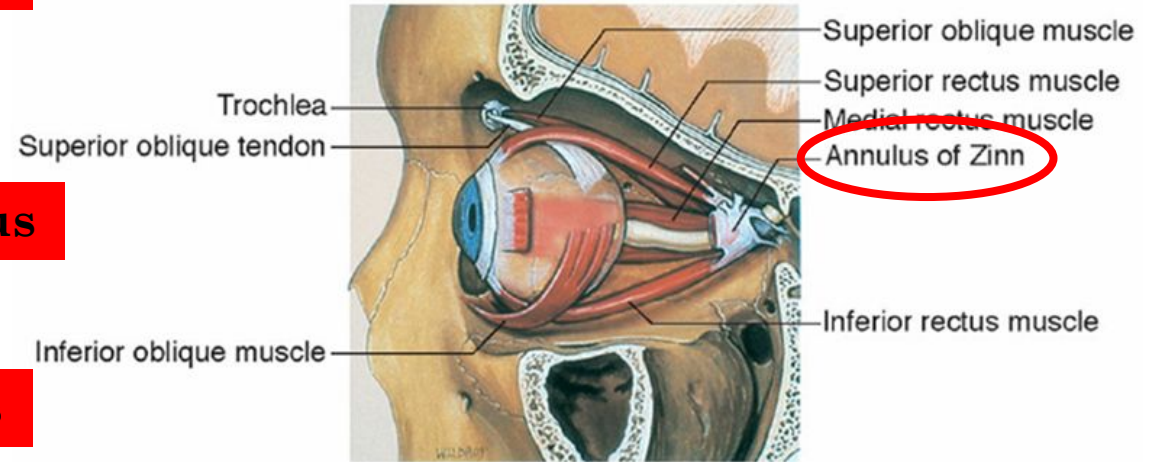
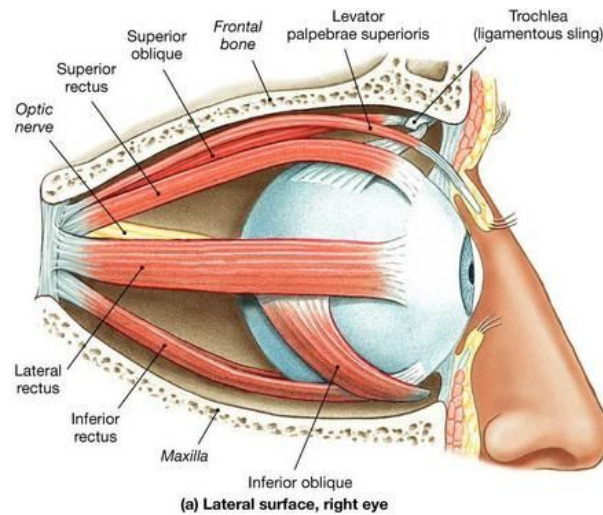
Lateral rectus

Medial rectus

Inferior oblique

Inferior rectus

Spiral of Tillaux



Extraocular muscles - กล้ามเนื้อตา

Table 3-1 Action of the Extraocular Muscles from Primary Position

Muscle*	Primary	Secondary	Tertiary
Medial rectus	Adduction	—	—
Lateral rectus	Abduction	—	—
Inferior rectus	Depression	Extorsion	Adduction
Superior rectus	Elevation	Intorsion	Adduction
Inferior oblique	Extorsion	Elevation	Abduction
Superior oblique	Intorsion	Depression	Abduction

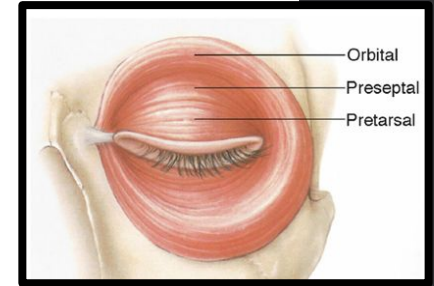
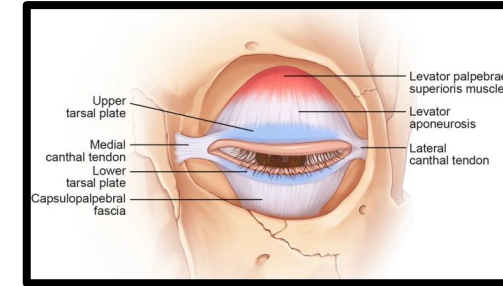
*The superior muscles are intortors; the inferior muscles, extortors. The vertical rectus muscles are adductors; the oblique muscles, abductors.



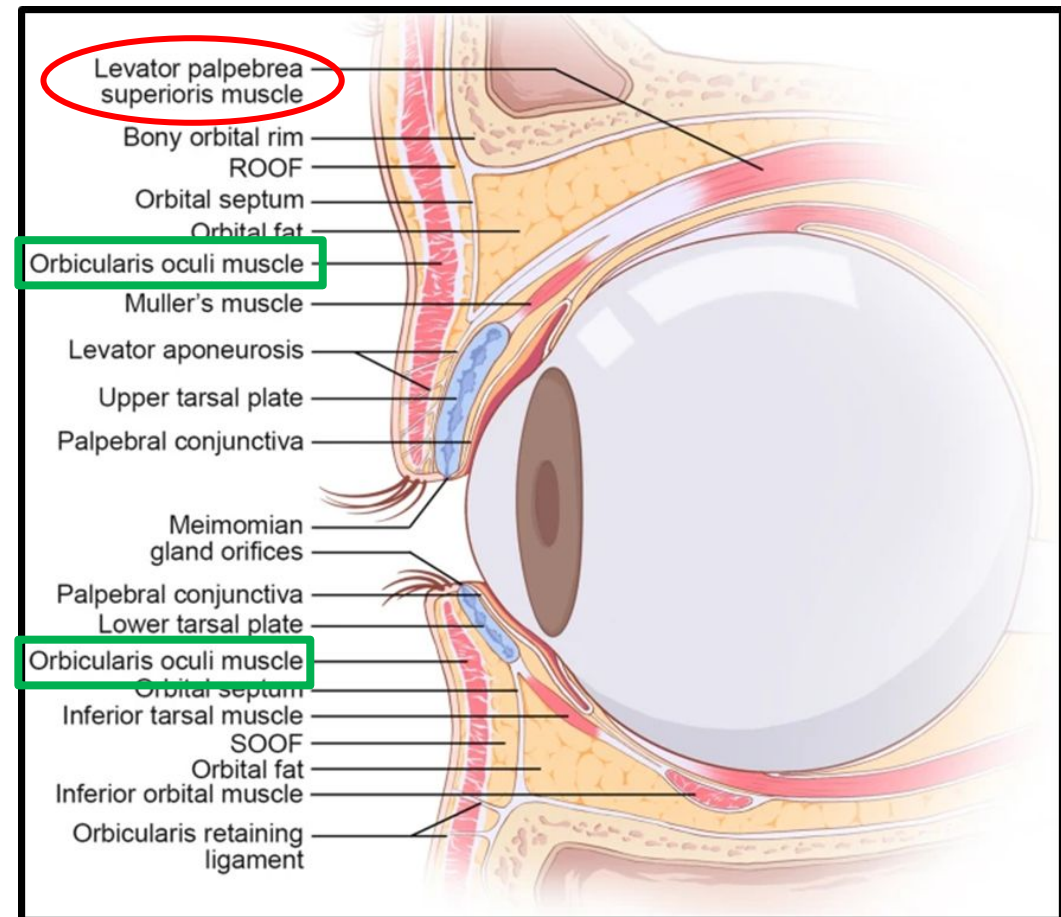
Innervation

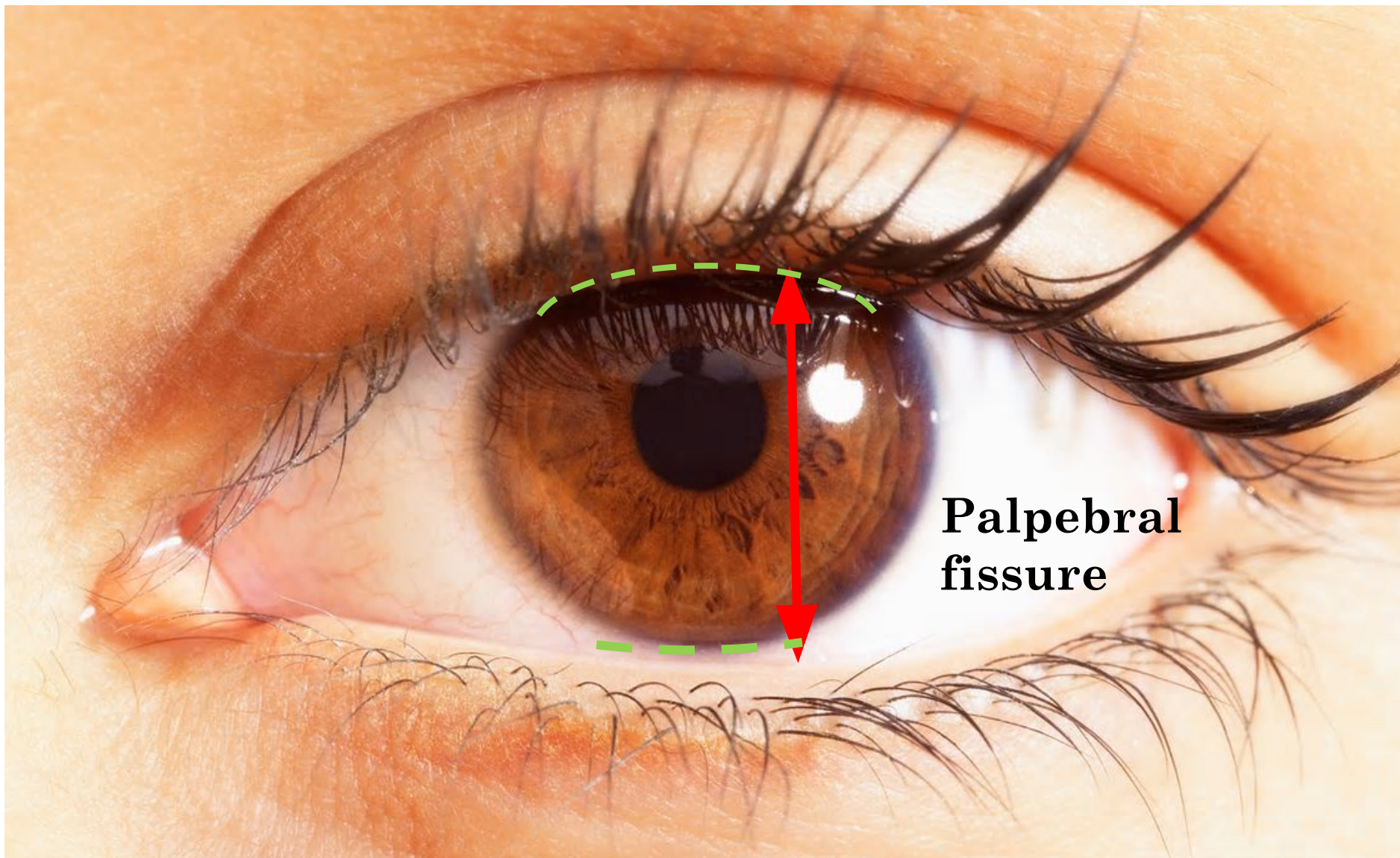
- SO : **CN IV**
- LR : **CN VI**
- MR, SR, IR, IO and levator palpebrae superioris : **CN III**

Eyelid - เปลือกตา



1. skin
2. eyelid margin
3. orbicularis oculi muscle
4. orbital septum
5. levator palpebrae superioris muscle
6. Müller muscle
7. tarsus

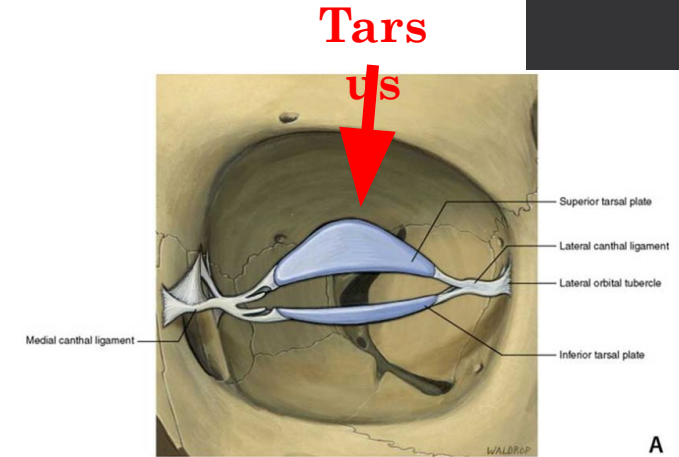
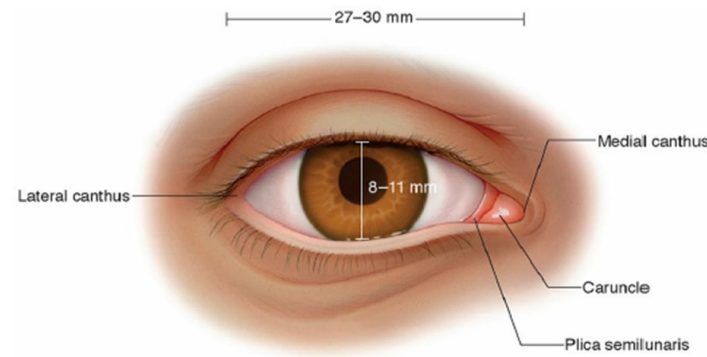




**Palpebral
fissure**

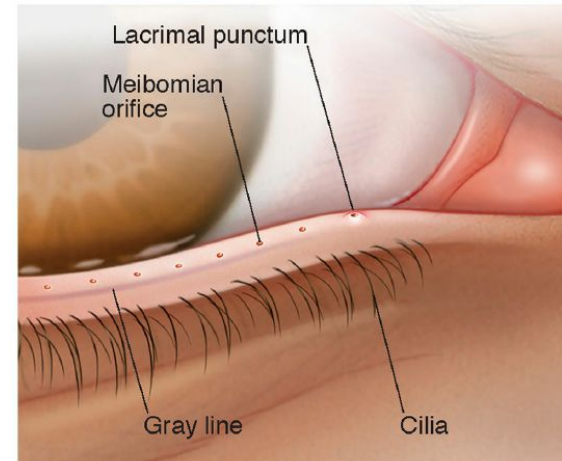
Eyelid skin

- The thinnest skin
- Contain of fine hairs, sebaceous glands, and sweat glands
- Superior eyelid fold = lid crease (insertional attachment of levator aponeurosis approximately 3 mm superior to the eyelid margin)



Eyelid margin

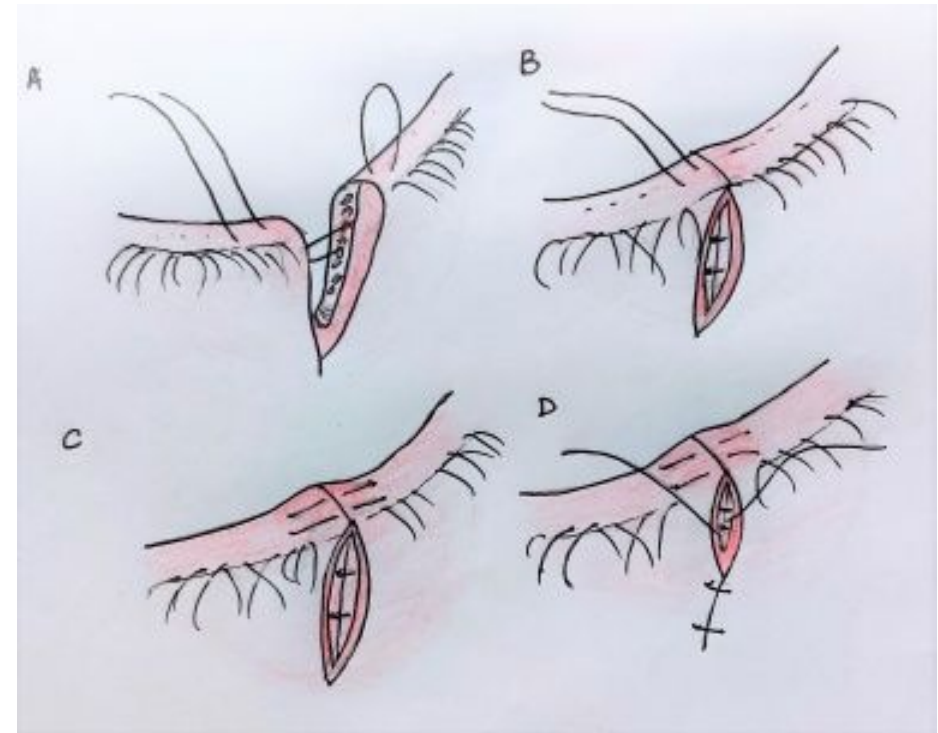
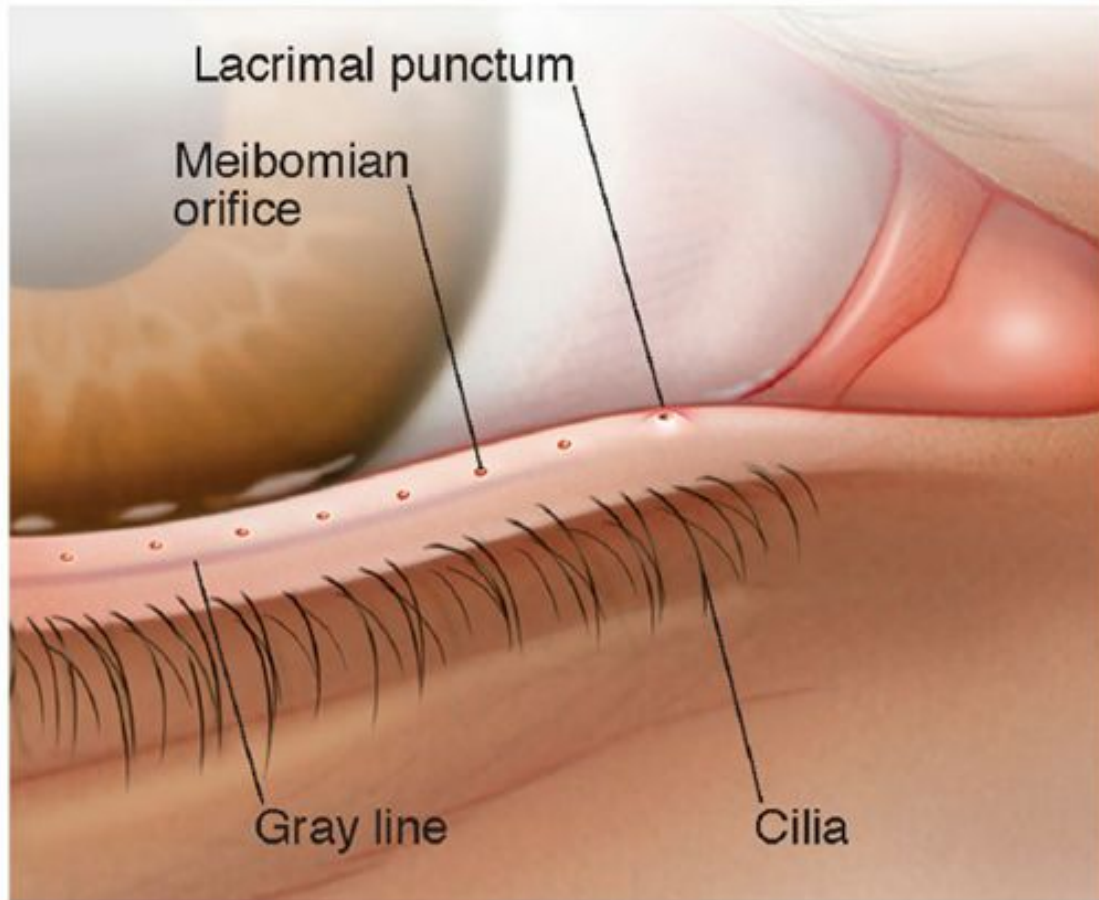
- The punctum of the canaliculus (superior&inferior) : medial aspect and meibomian gland orifice
- Gray line (or intermarginal sulcus)
- Eyelashes (or cilia) : 2 or 3 irregular rows along the anterior dermal edge of the eyelid margin



The Form and Function of Meibomian Glands

Mark B. Abelson, MD, CM, FRCSC, FRCO, George Ousler, Aron Shapiro and David Rimmer, Andover, Mass.

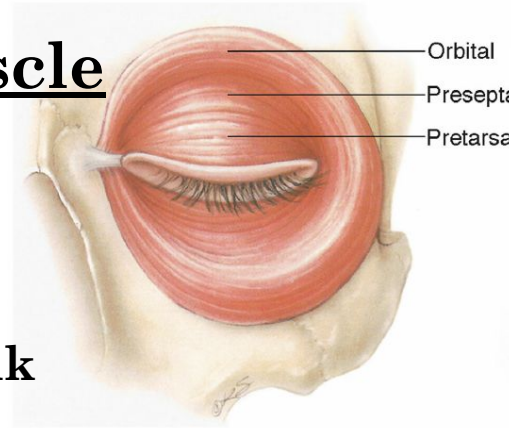
Eyelid margin



Eyelid reconstruction

Orbicularis oculi muscle

- orbital part
- preseptal part
- pretarsal part
- Innervation : CN VII
- Function : **close eyelid, blink**

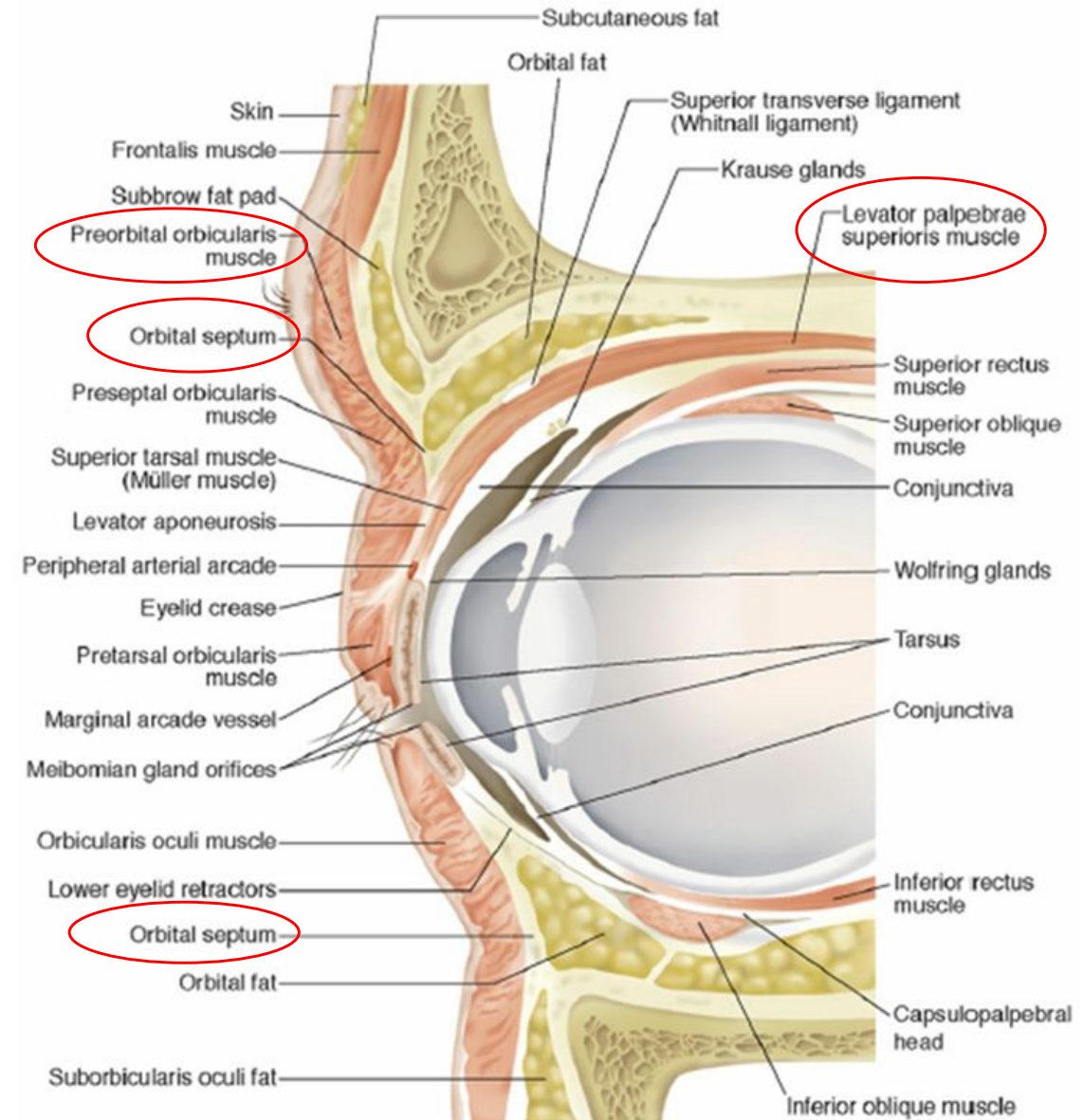


Orbital septum

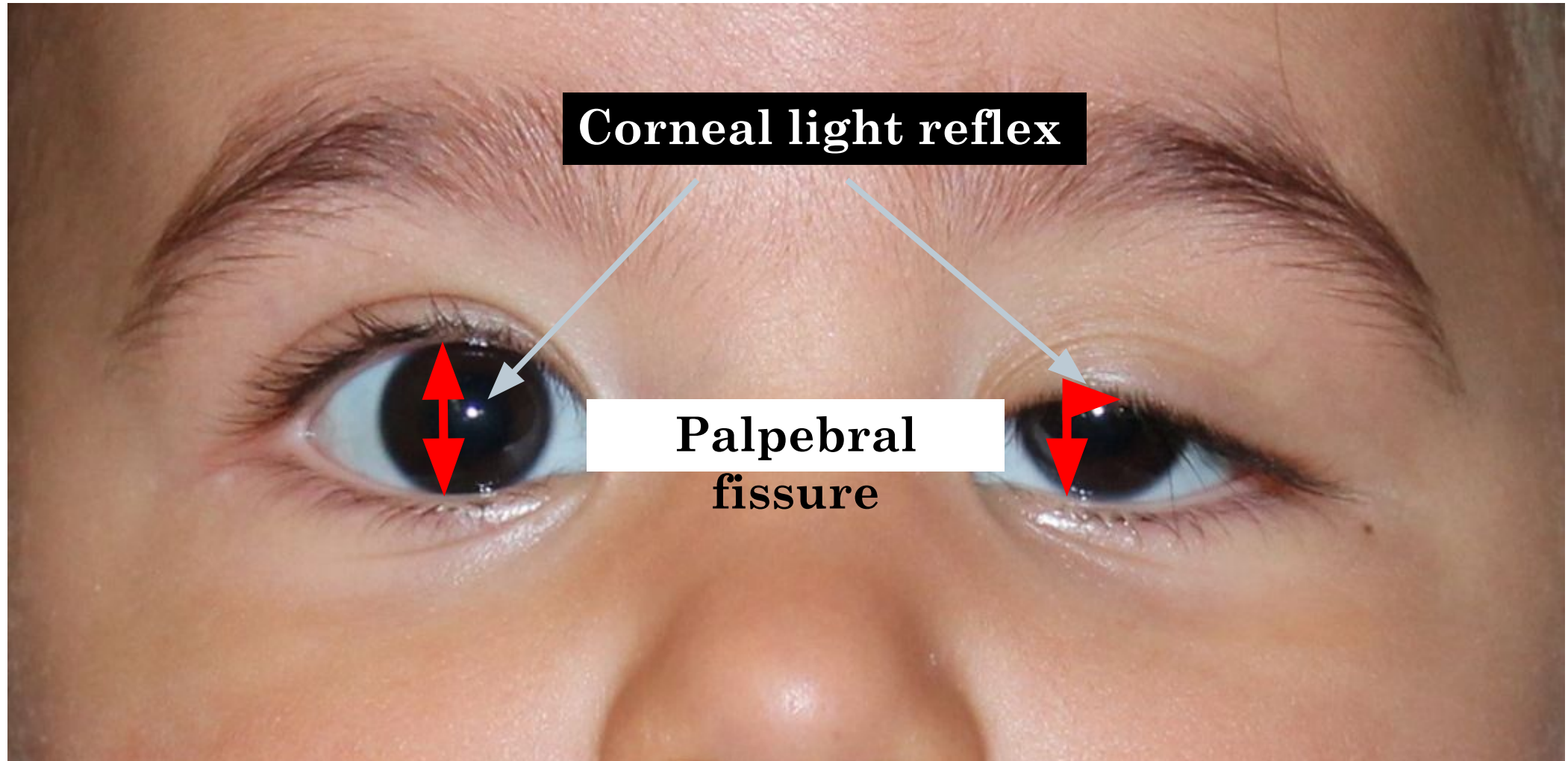
- A thin sheet of connective tissue that encircles the orbit as an extension of the periosteum of the roof and the floor of the orbit and attaches to the anterior surface of the levator palpebrae superioris muscle
- Provide a barrier to anterior or posterior extravasation of blood or the spread of inflammation

Levator palpebrae superioris

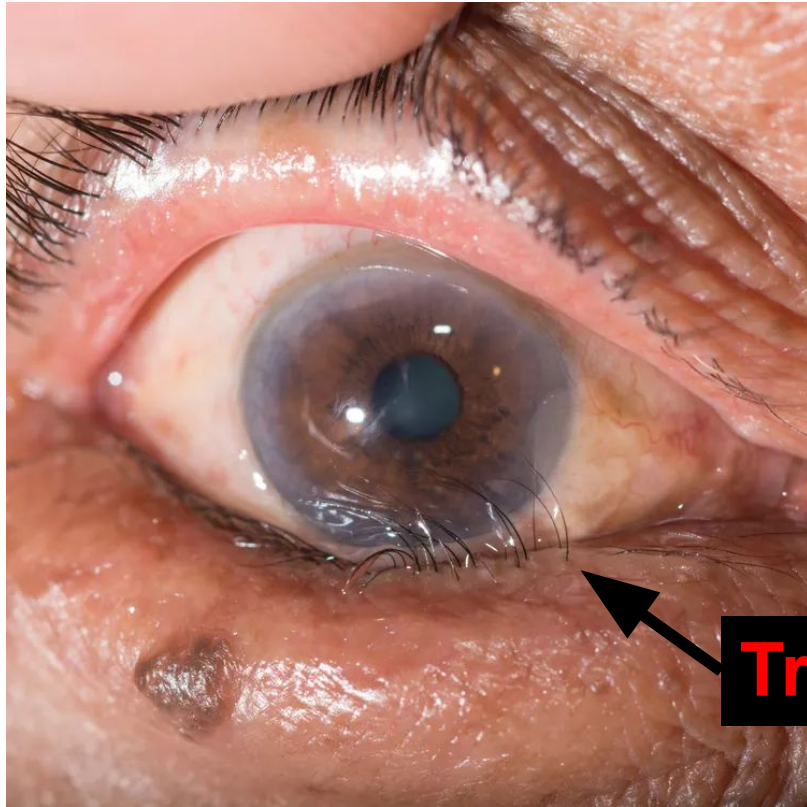
- Function : open eyelid
- Innervation : CN III



Ptosis หนังตาดตก



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Vijaya Nethralaya



Trichiasis

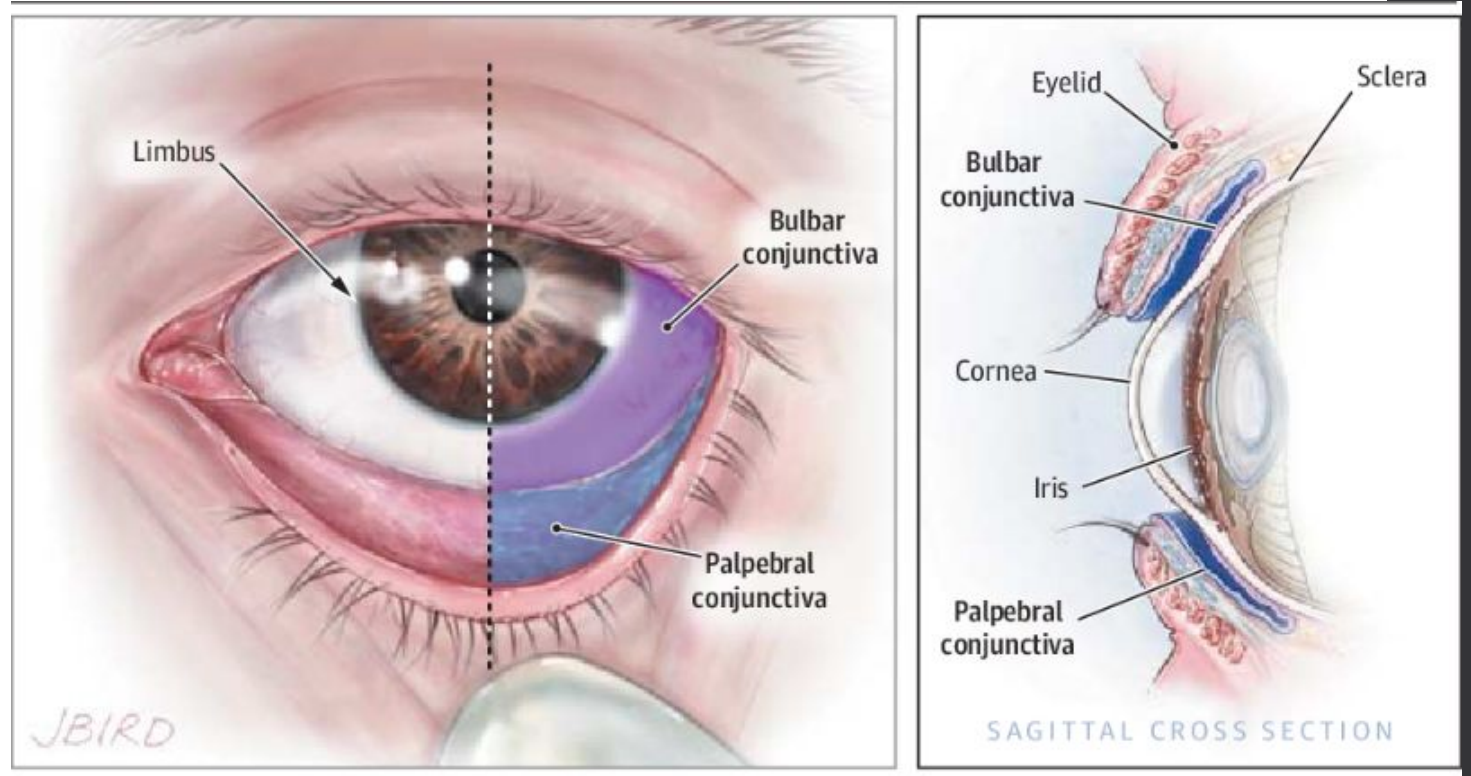
Entropion



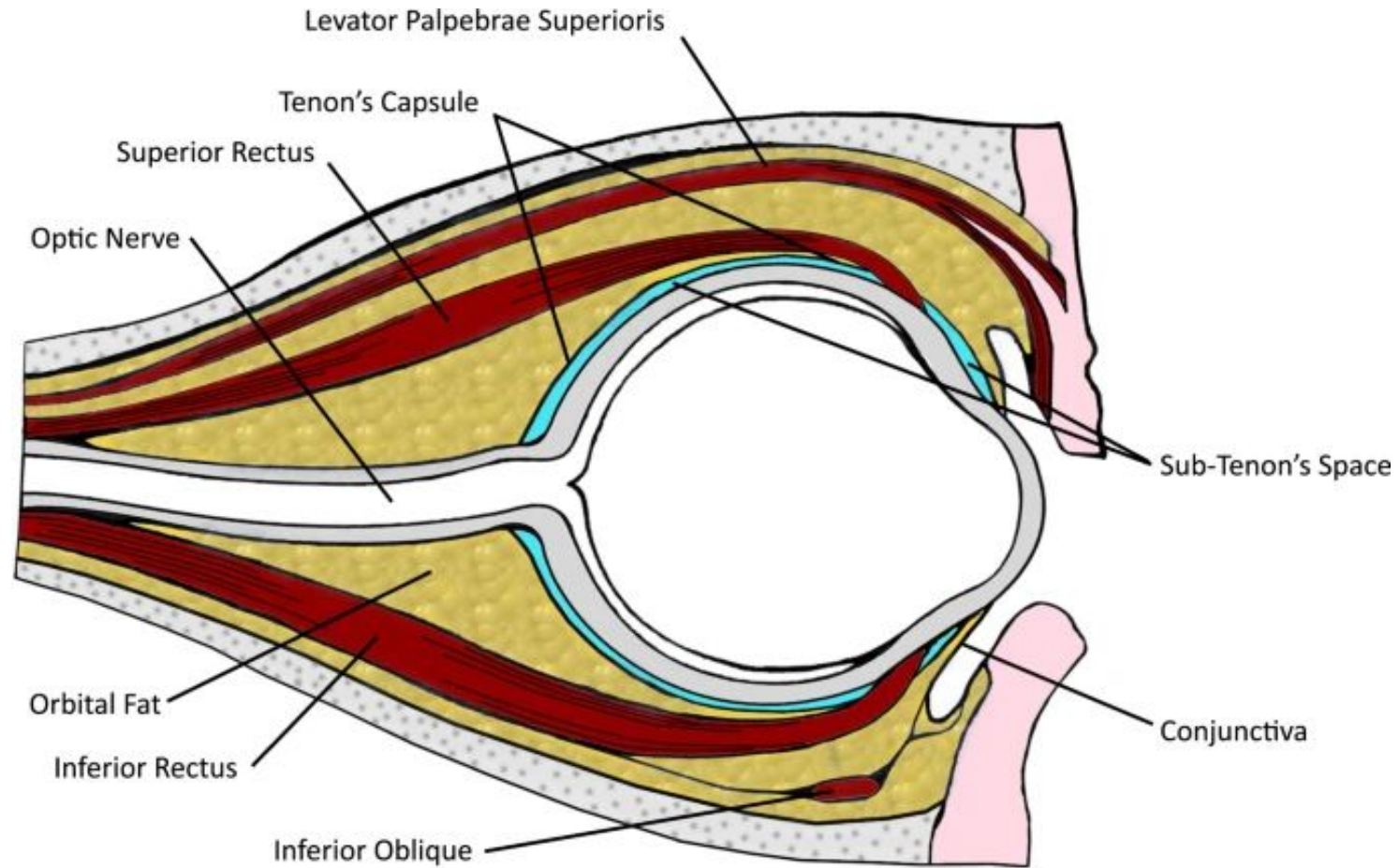
Ectropion

Conjunctiva - เยื่อบุตา

- Transparent vascularized membrane covered by a nonkeratinized epithelium
- 3 parts : palpebral, forniceal, and bulbar
- Globlet cells produce mucin (tear)



Tenon's capsule



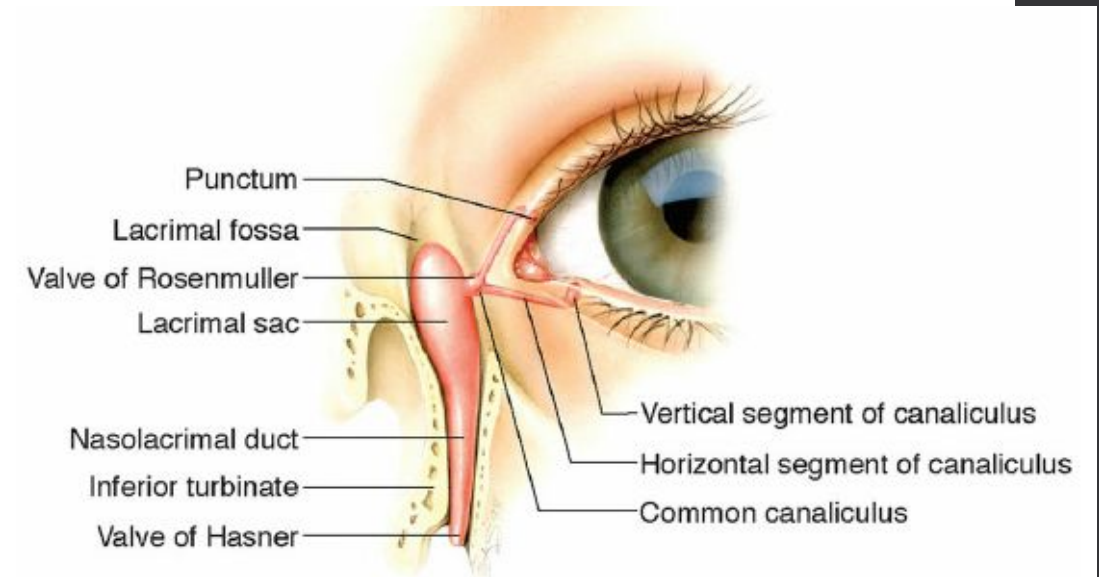
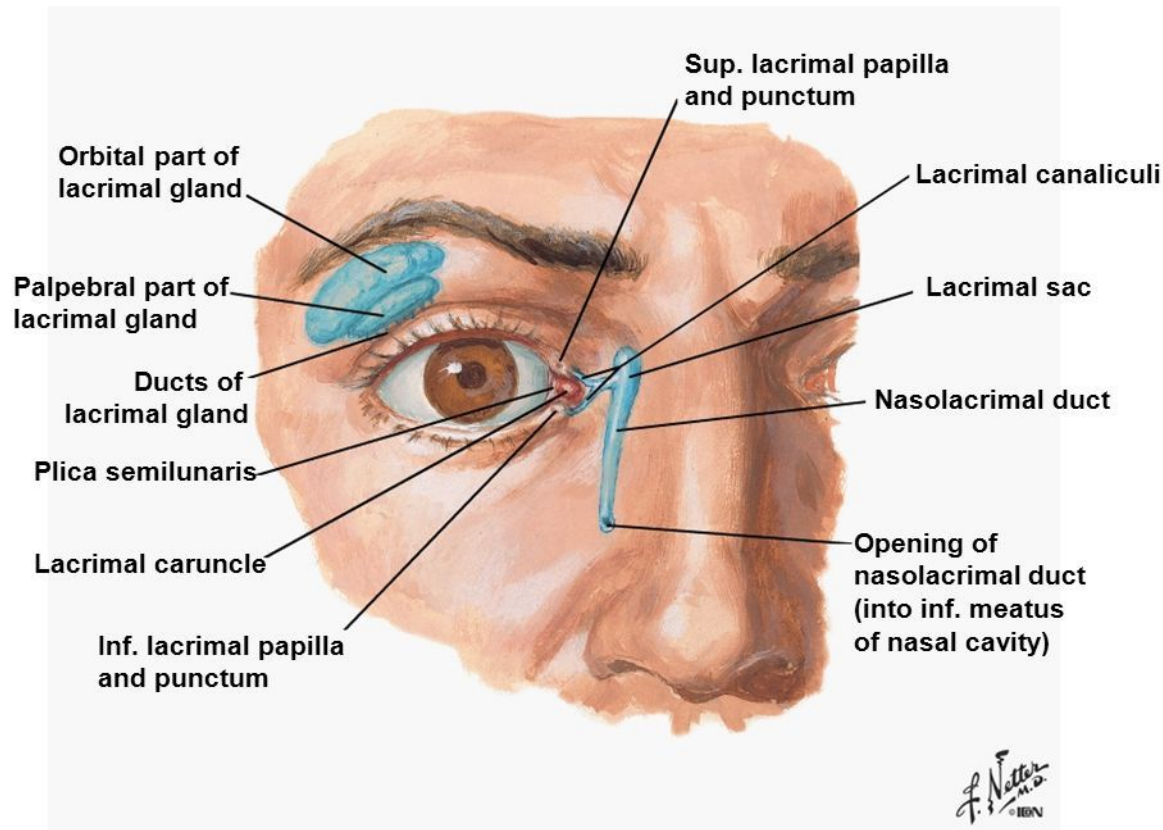
Chua, M.J., Lersch, F., Chua, A.W.Y. et al. Sub-Tenon's anaesthesia for modern eye surgery—clinicians' perspective, 30 years after re-introduction. *Eye* 35, 1295–1304 (2021). <https://doi.org/10.1038/s41433-021-01412-5>

Lacrimal system - ระบบท่อ น้ำตา

Lacrimal gland

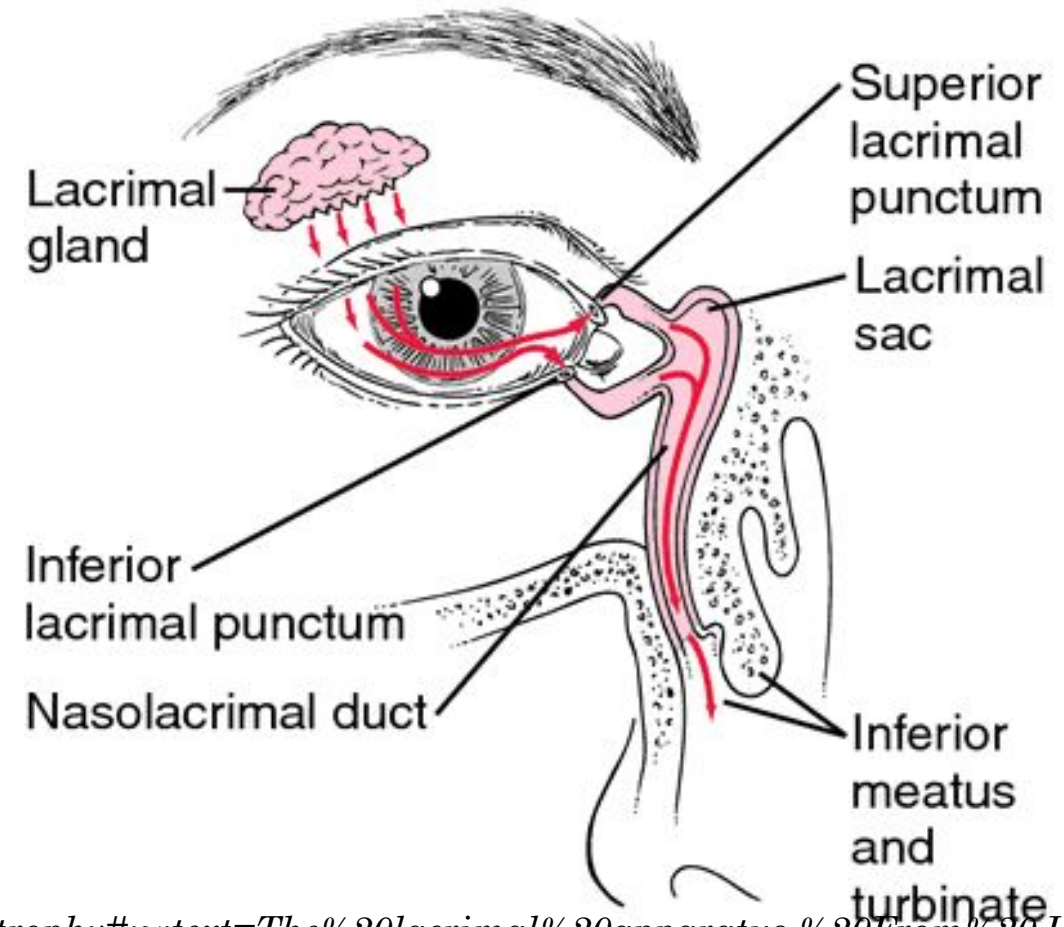
- Main orbital part
- Palpebral part
- Lacrimal gland **biopsy** should be performed on the main part to avoid sacrificing the ducts

LACRIMAL APPARATUS



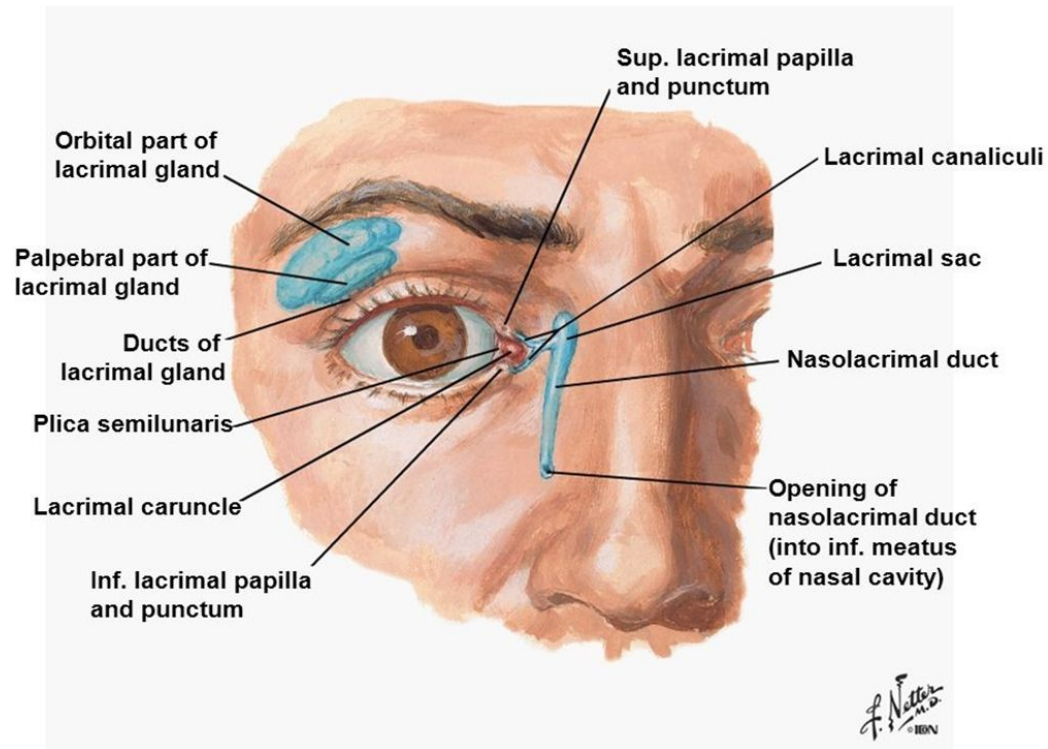
Tear secretion

- **Basic tear**
- **Reflex tear**
 - Foreign bodies
 - Chemical
 - Emotional
 - Light



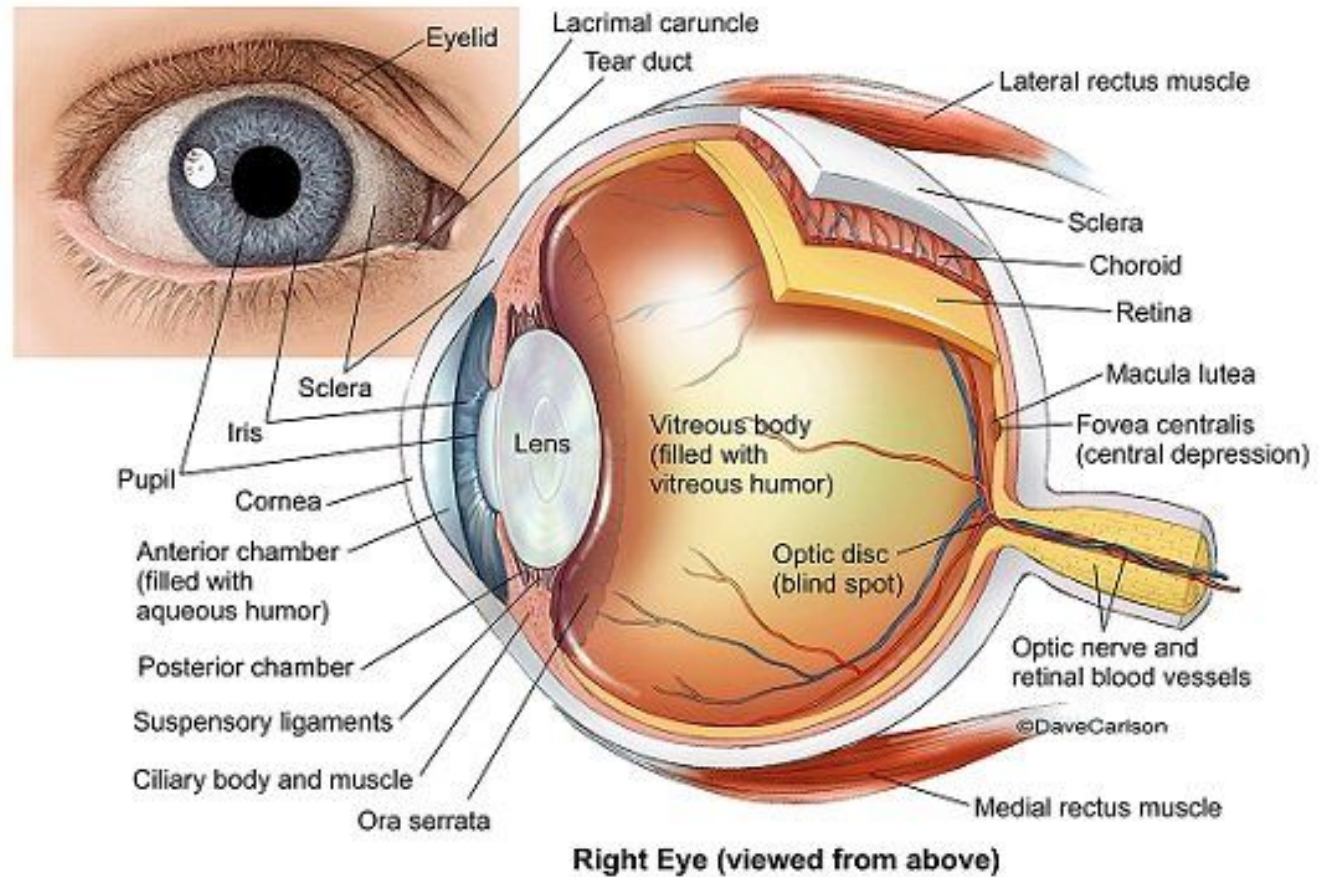
Canaliculi injury

LACRIMAL APPARATUS



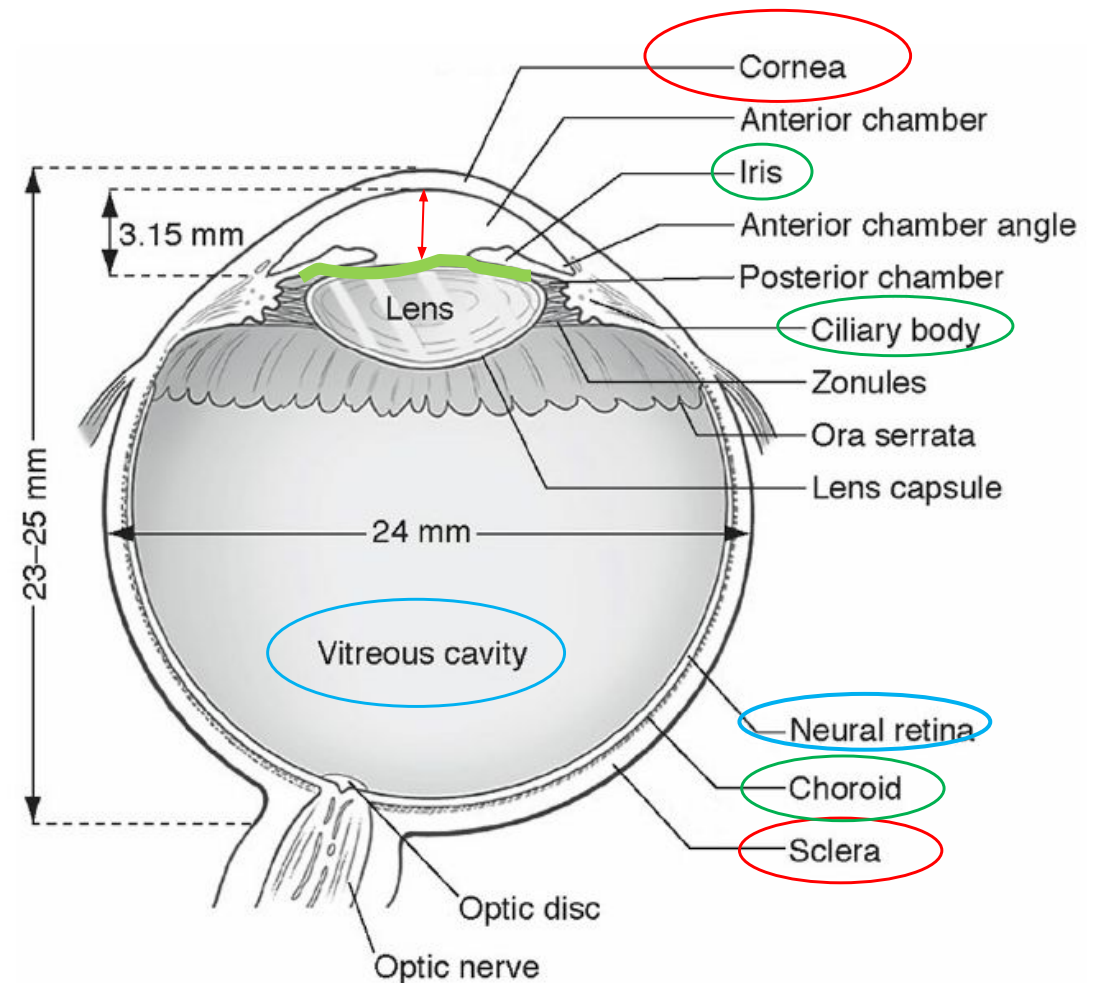
Eye ball - ลูกตา

- Cornea
- Sclera
- Lens
- Aqueous outflow apparatus
- Uvea
 - Iris, ciliary body and choroid
- Vitreous
- Retina
- Optic nerve

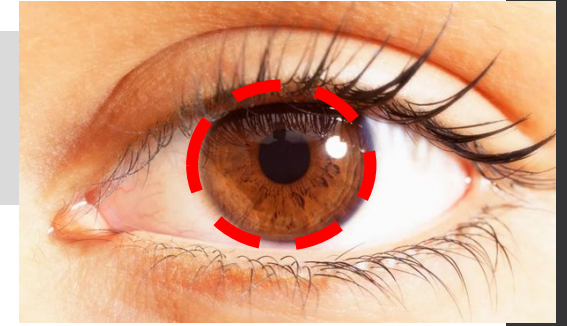


Eyeball

- **Axial length (AL)** is approximately **23–25 mm**
 - Myopic eyes : long AL
 - Hyperopic eyes : short AL
- 3 compartments
 - Anterior chamber
 - space between the iris and the cornea
 - Posterior chamber
 - space between the iris and the lens
 - Vitreous cavity
- 3 layers
 - Outer layer : cornea and sclera
 - Middle layer : uvea (iris, ciliary body and choroid)
 - Inner layer : retina

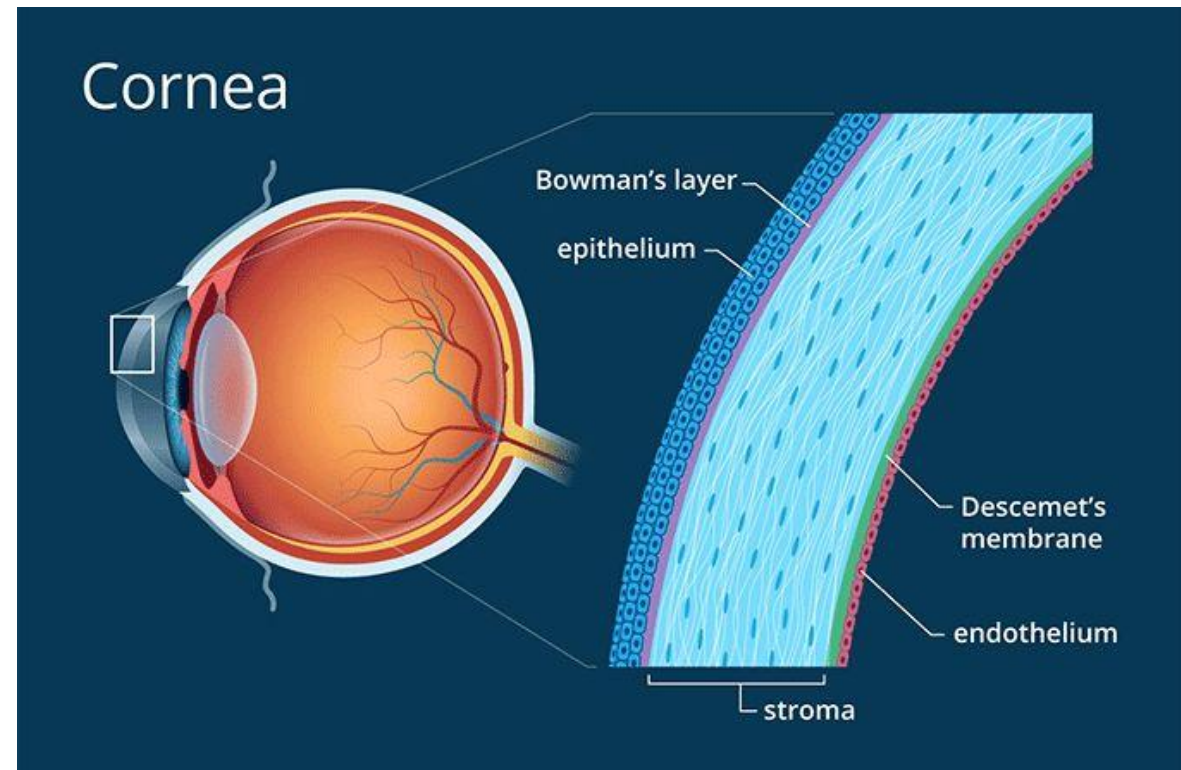


Cornea - กระฉกตา



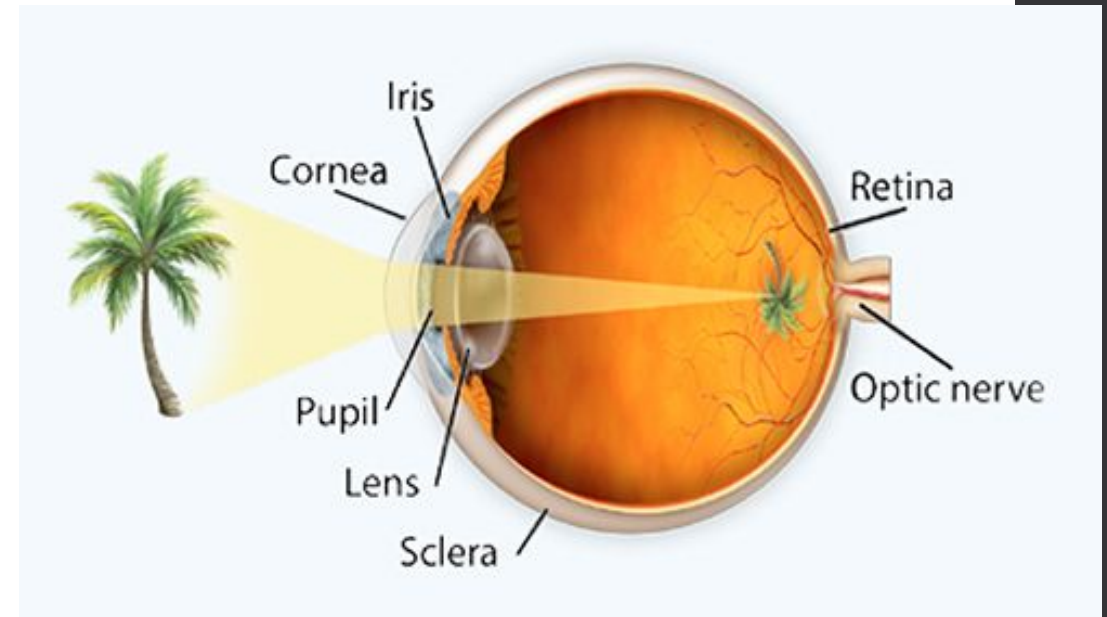
Limbus

- corneal diameter
 - Horizontal $\varnothing$ = 12 mm
 - Vertical $\varnothing$ = 11 mm
- Refractive power = 43 diopters (D)
- Central corneal thickness $\sim$ 0.5mm
- Consists of five layers
 - Epithelium
 - Bowman's layer
 - Stroma
 - Descemet membrane
 - Endothelium



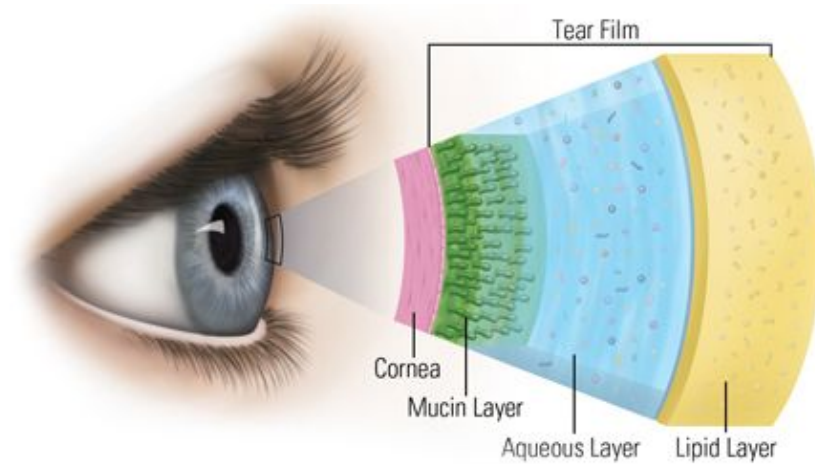
Cornea - กระพจกตา

- Avascular and transparent structure
- Function : barrier, protection and refraction of light
- Corneal nutrition and metabolism
 - oxygen
 - tear, aqueous humor, limbal vessels
 - glucose and other nutrition
 - aqueous humor



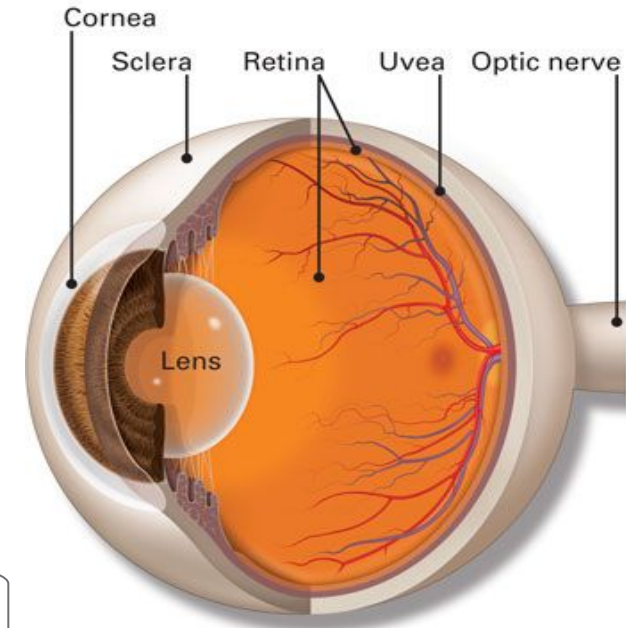
Precorneal tear film

- 3 layers (3-4 microns)
 1. *Lipid layer*
 2. *Aqueous layer*
 3. *Mucin layer*
- Function
 - **Lubricating** the surface of the cornea and conjunctiva
 - **Diffusion of oxygen** and other nutrients
 - **Prevent infection** : contain immunoglobulins, lysozyme, and lactoferrin
 - **Prevents the evaporation of tears** from the surface of the eye
 - **Maintains surface tension**



Sclera - ตาขาว

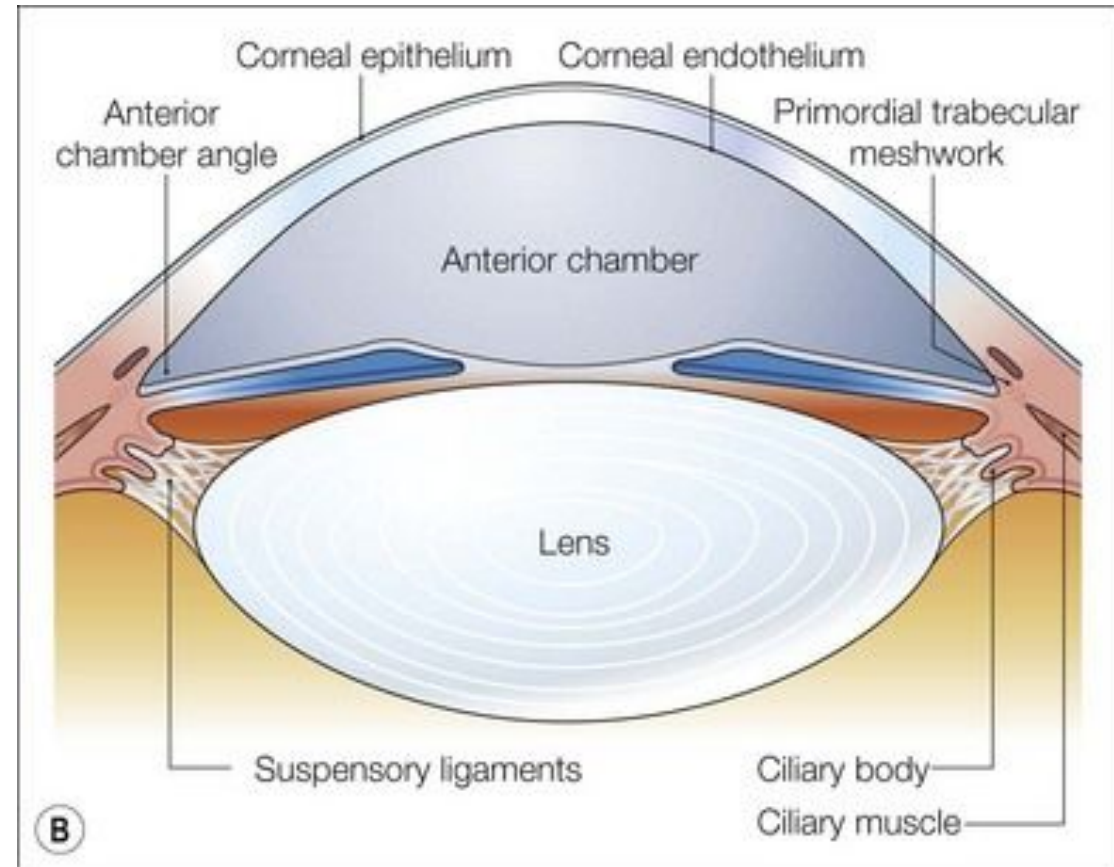
- Anterior opening = cornea, posterior opening = optic nerve
- Muscle insertion
- Covered by tenon's capsule
- ***Thinnest part*** (0.3 mm) : just posterior to muscle insertion
- Scleral rupture following blunt trauma can occur at several sites:
 - corneal limbus opposite the site of impact
 - the insertion of the rectus muscles
 - the equator of the globe
 - The most common site is the superonasal quadrant near the limbus



**Weak
point**

Anterior chamber - ช่องหน้าม่านตา

- The depth of the anterior chamber is approximately 3 mm deep at its center.
- The volume is approximately 200 μL .
- Filled with aqueous humor, which is ***produced***
- ***by ciliary body*** in the posterior chamber

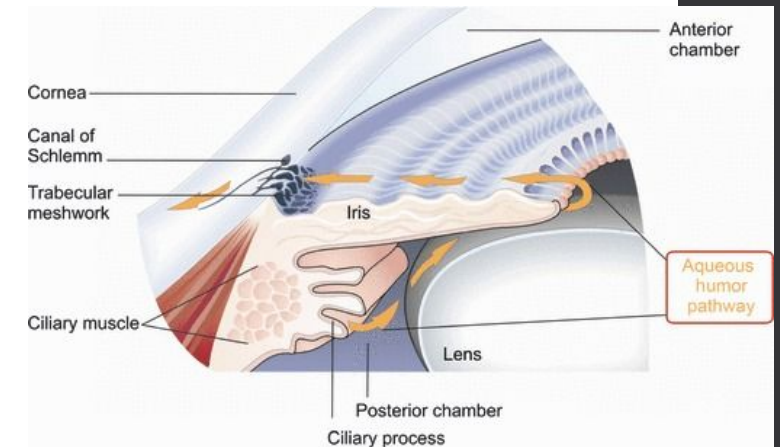
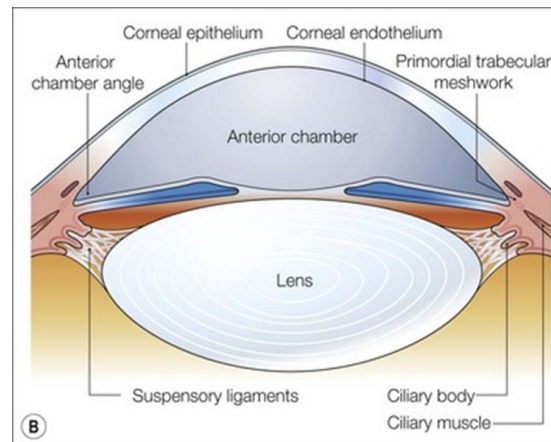


Anterior segment: developmental anomalies |
Ento Key

Aqueous humor

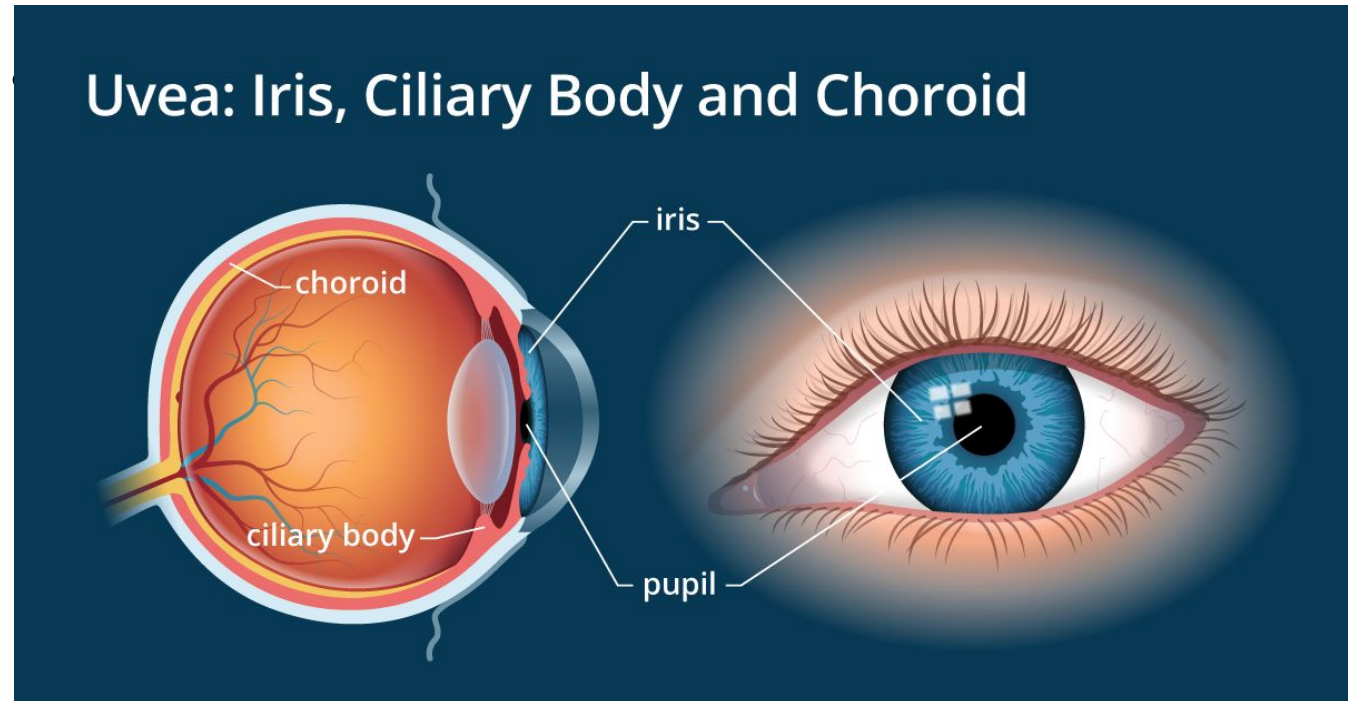
- Pathway of aqueous humor
 - **Posterior chamber -> pupil -> anterior chamber -> trabecular meshwork -> Schlemm canal**
 - Partly by uveoscleral pathway drainage into the supraciliary space
- Functions of aqueous humor
 1. nutrition and oxygen source for cornea, lens and trabecular meshwork
 2. remove metabolic waste products
 3. maintain appropriate IOP
 4. optically clear medium

IOP (intraocular pressure) 10-16 mmHg



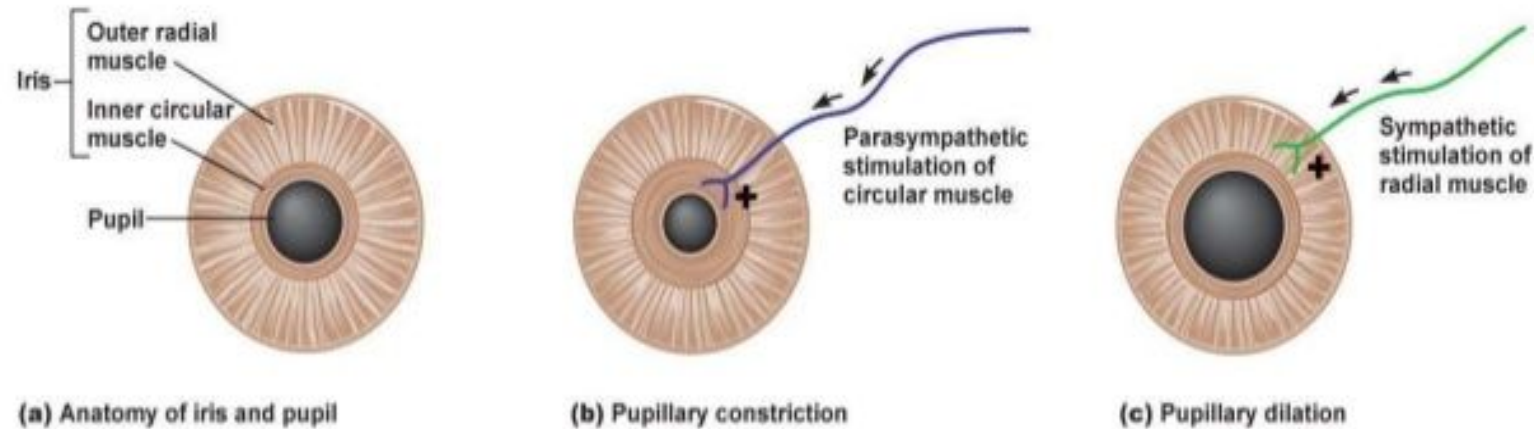
Uvea (uveal tissue)

- Uveal tissue consists of parts
 - Iris
 - Ciliary body
 - Choroid



Iris and Uvea of the Eye - All About Vision

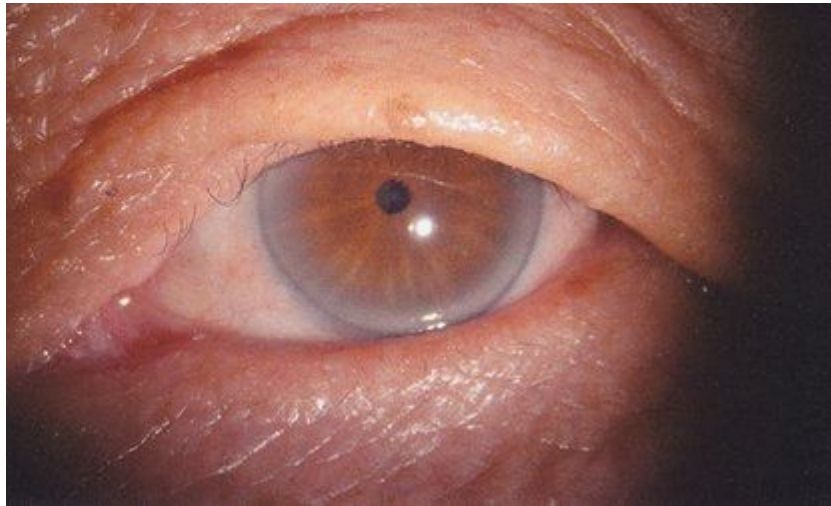
Iris - ม่านตา



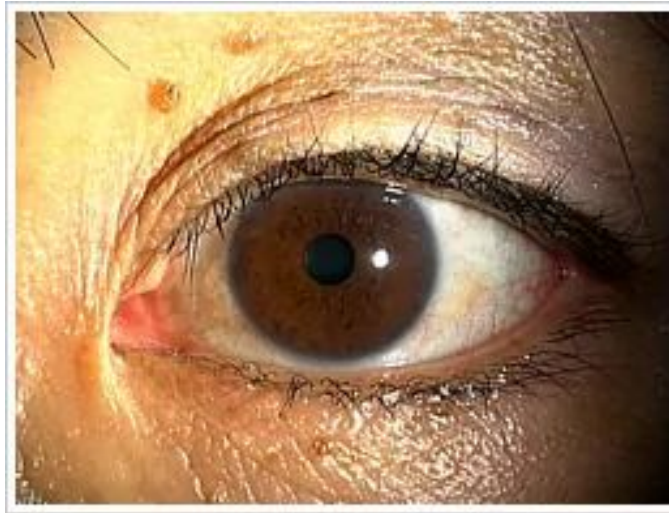
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- The **most anterior** extension of the uveal tract
- Consist of **connective tissue, blood vessels, melanocytes + pigment cells, dilator muscle, sphincter muscle**
- Mobility of the iris allows the pupil to change size: mydriasis vs miosis
- Between anterior – posterior chamber

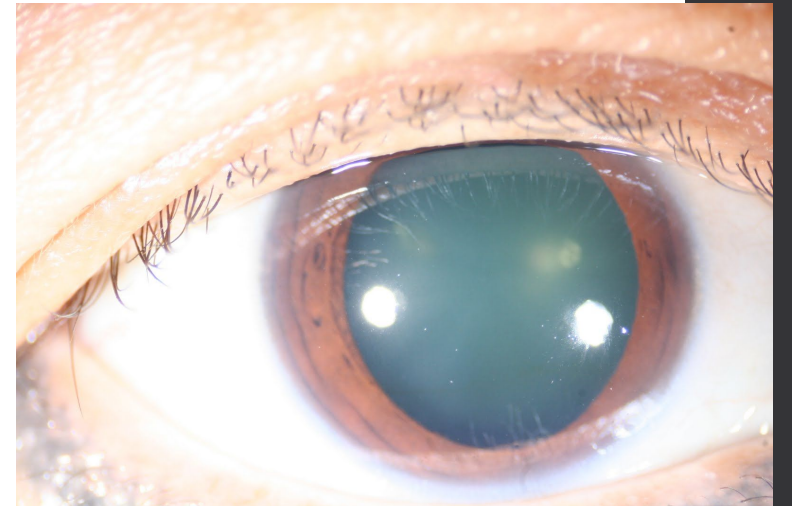
Pupil รูม่านตา



Miosis
s



Normal pupil
size



Mydriasis

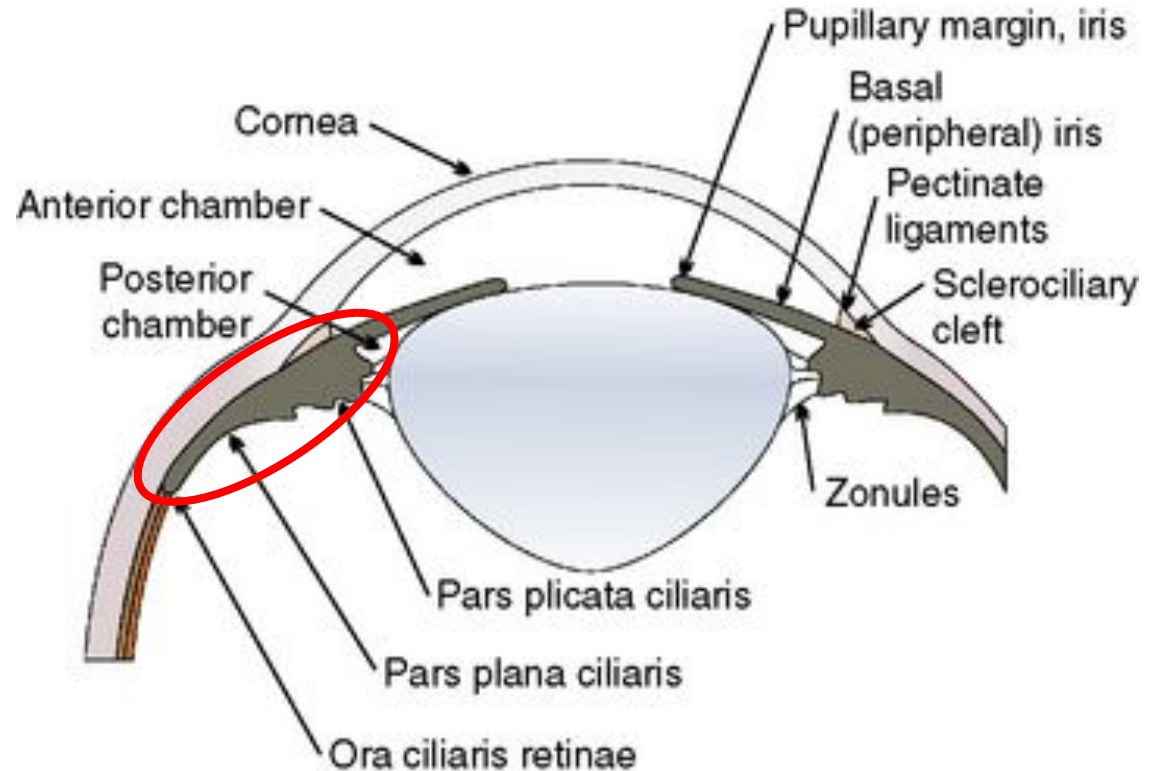
Ciliary body

- Functions

- aqueous humor formation
- aqueous humor drainage
- lens accommodation

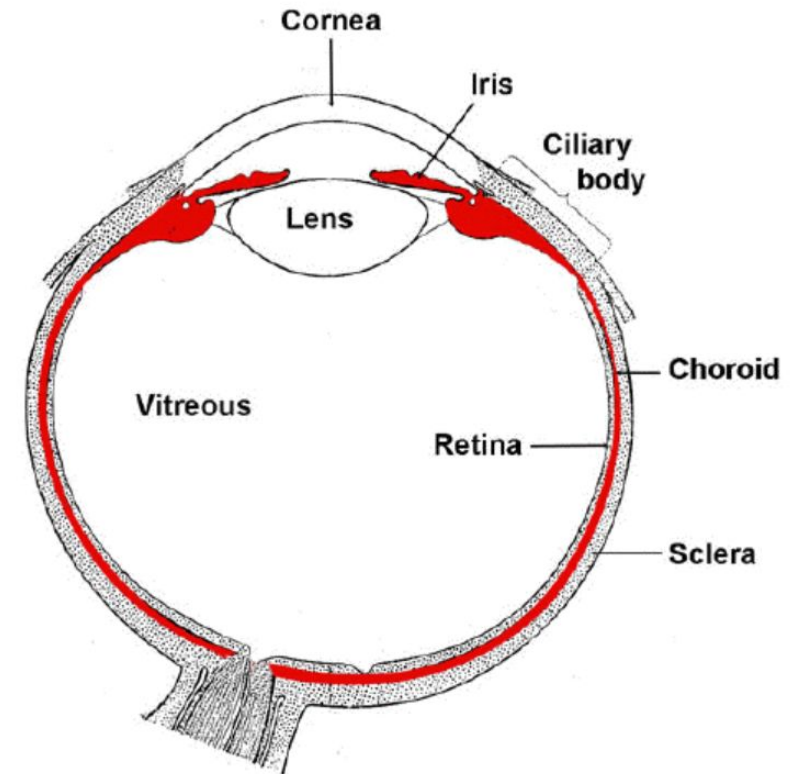
- Ciliary muscle

- Longitudinal
- Radial
- Circular



Choroid

- **Posterior portion** of uveal tract
- **Nourishes the outer portion of the retina**
- **Consists of 3 layers of vessels:**
 - the choriocapillaris, the innermost layer
 - a middle layer of small vessels
 - an outer layer of large vessels



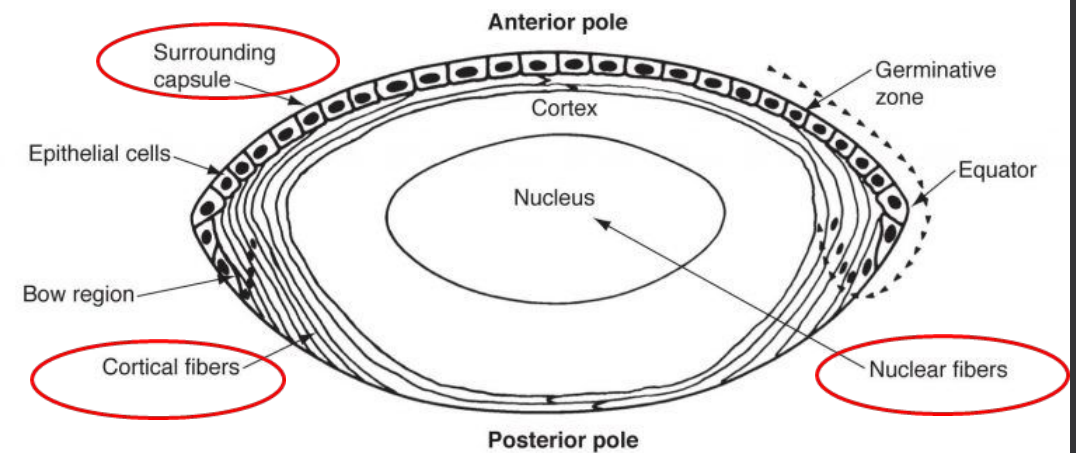
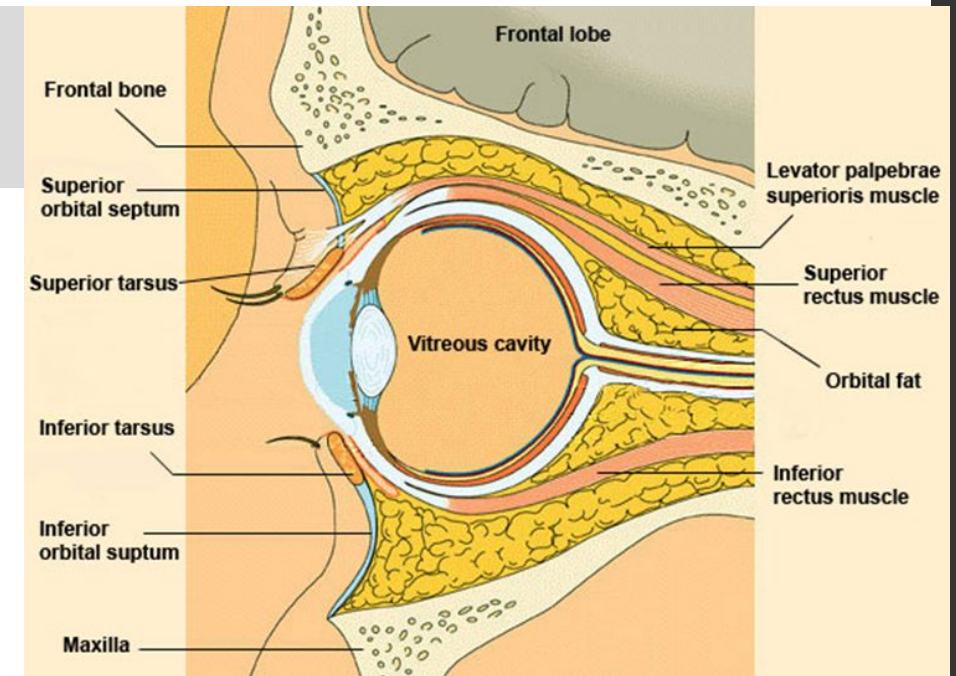
PHOTOBIOLOGY of the UVEAL TRACT

Dan-Ning Hu

The New York Eye and Ear Infirmary, Tissue Culture Center
310 East 14th Street, New York, NY 10003
hu2095@yahoo.com

Lens

- **Transparency**
 - precise organization of lens fiber cells
 - no membrane-bound organelles
 - high concentration of crystallins
- **Refraction of light**
 - High refractive index
- **Accommodation**

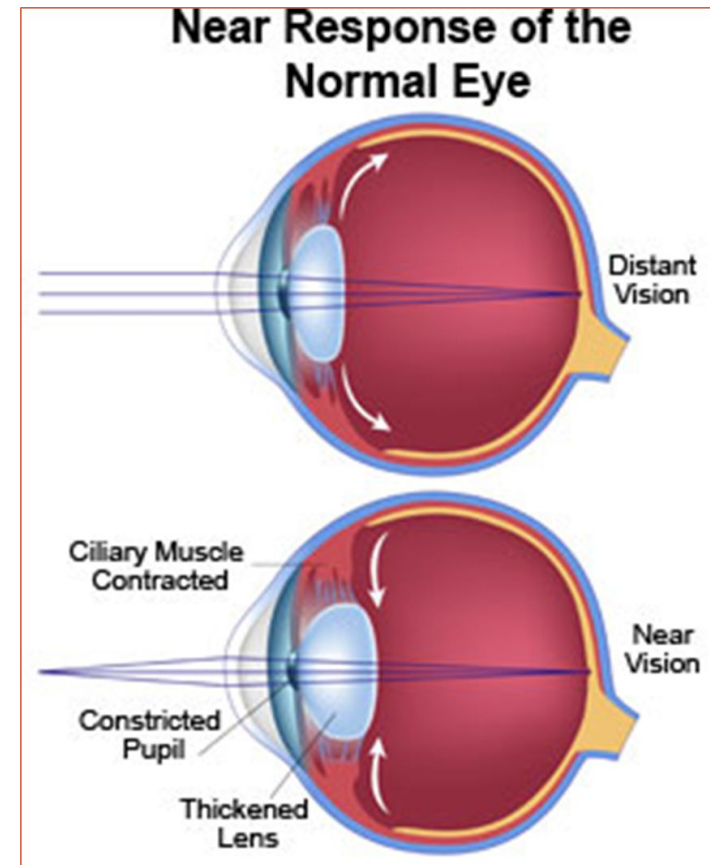


Accommodation

Ciliary muscle contraction

Zonular fiber relaxing

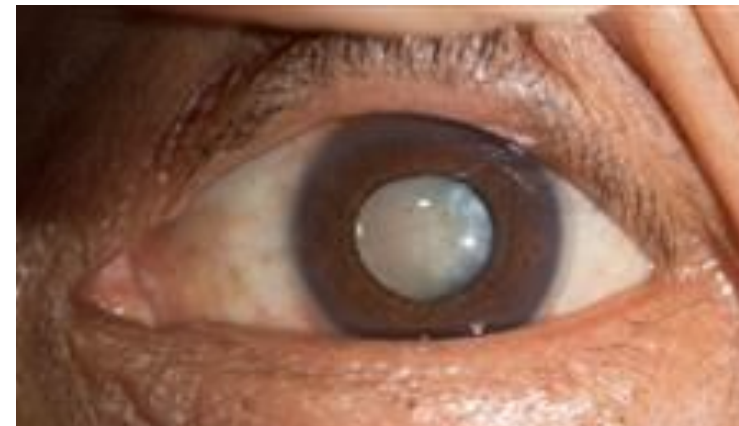
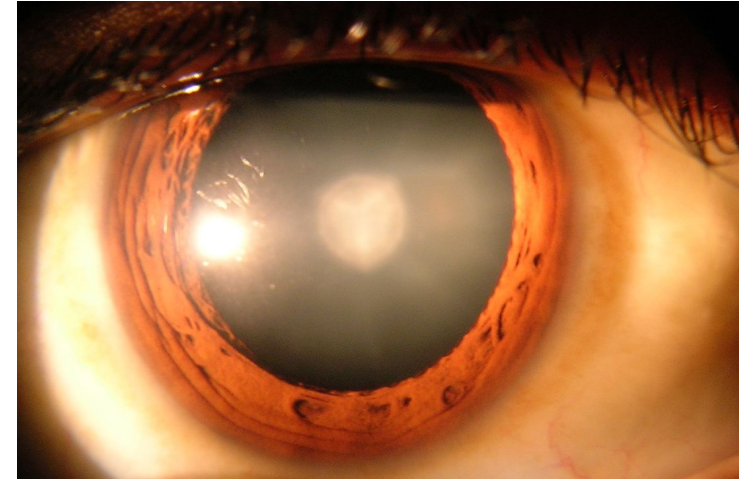
Lens become more convex

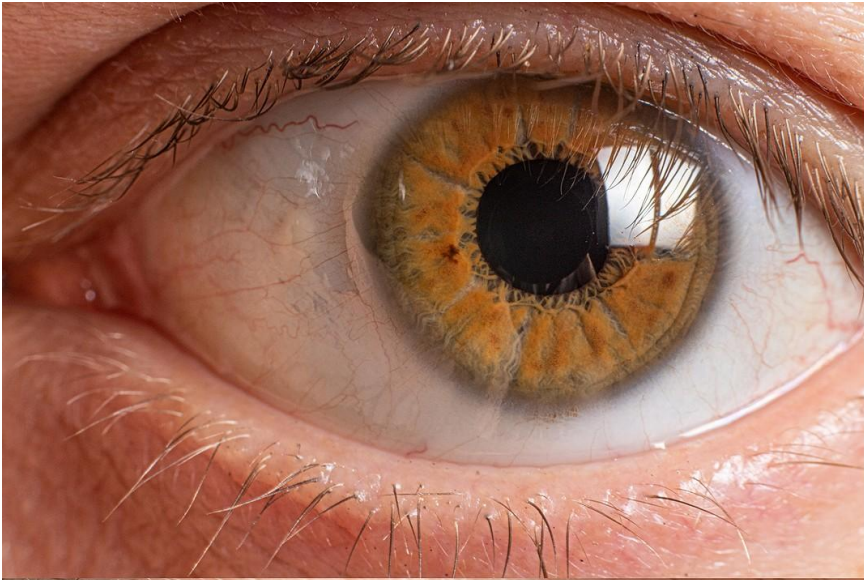


Near reflex : accommodation, convergence and miosis

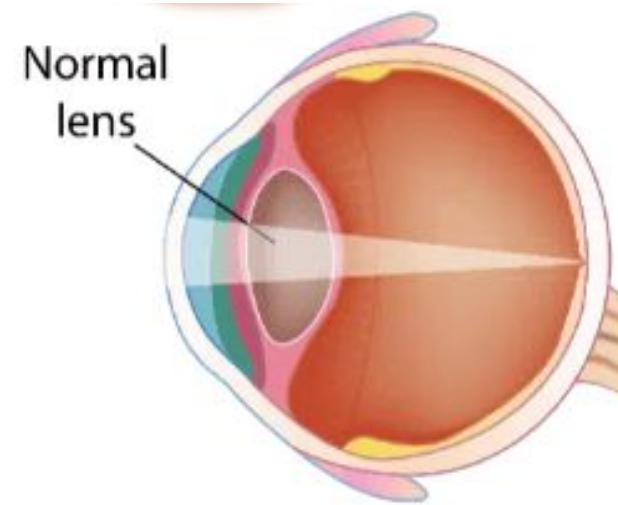
Cataract : lens opacity

- All nutrients must be obtained from aqueous humor and also released into aqueous humor
- **Cataractogenesis** : imbalance ionic homeostasis and protein in the lens
- **Causes**
 - Aging
 - Trauma
 - Congenital infection
 - Systemic diseases
 - Drug - steroid

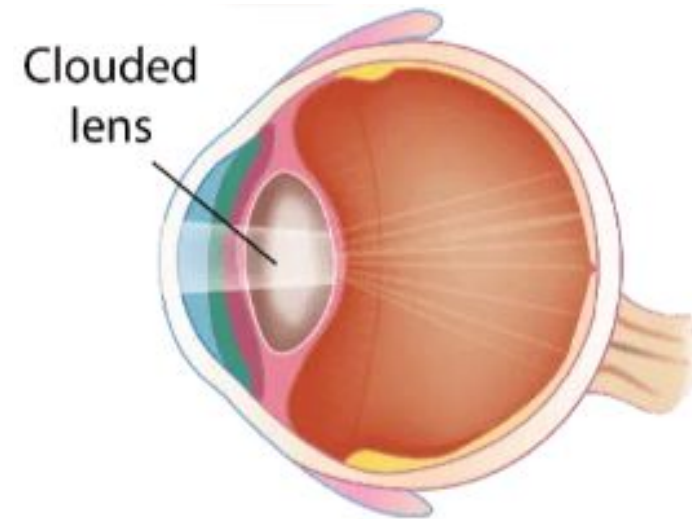




Normal
Eye



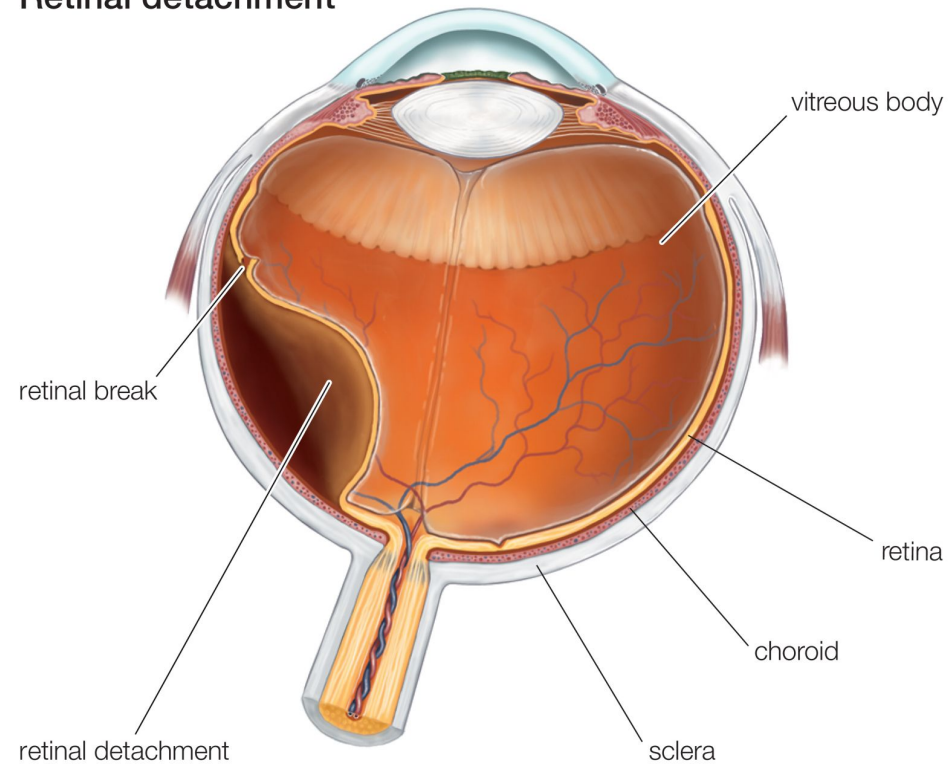
Cataract



Vitreous - น้ำวุ้นตา

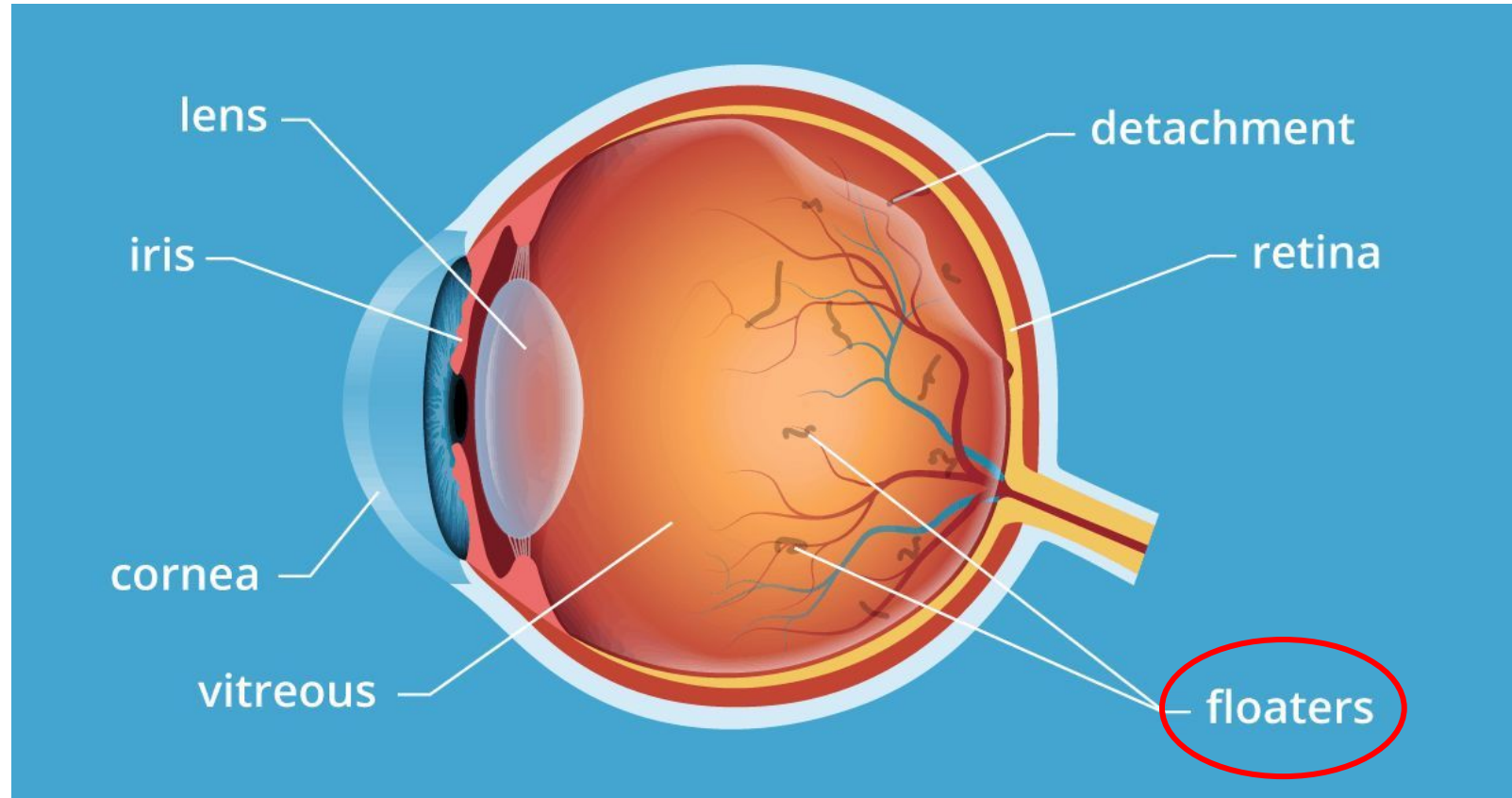
- volume ~ 4.0 ml.
- gel-like structure: 99% water, 1% hyaluronic acid and collagen
- adheres to the retina at
 - vitreous base
 - disc margin
 - perimacular region
 - along the retinal vessels
 - periphery of the posterior lens capsule

Retinal detachment



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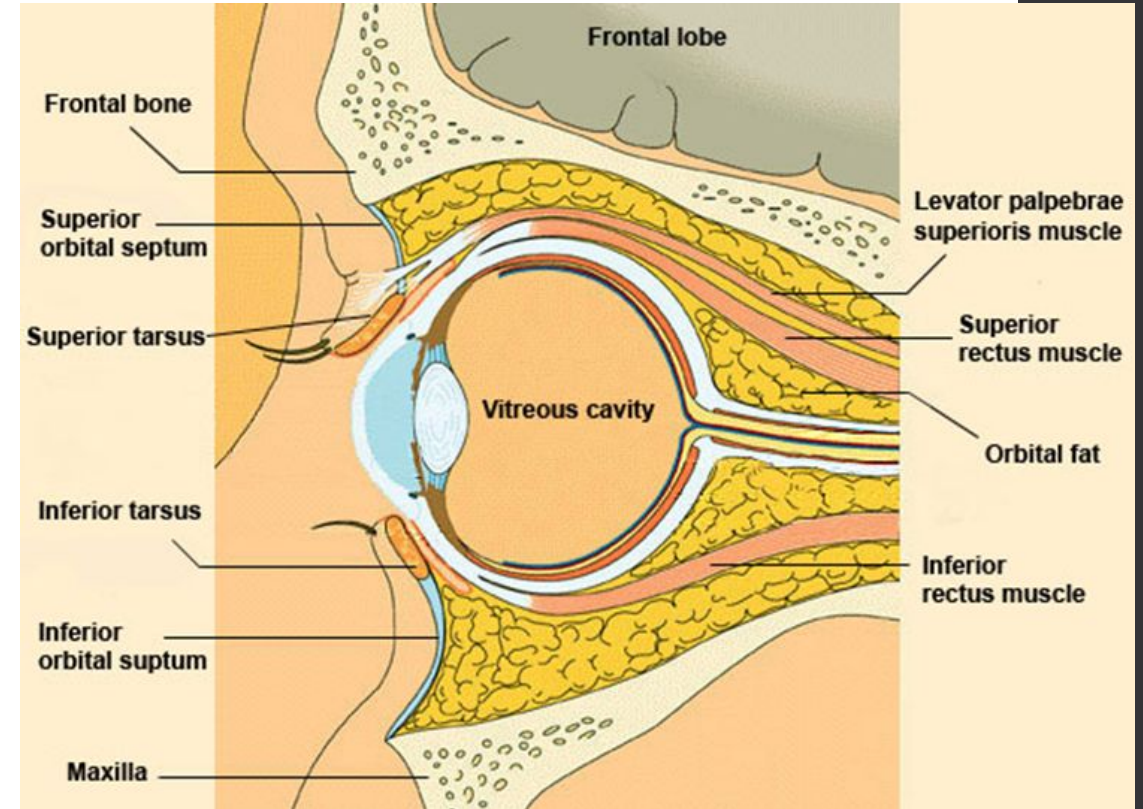
Posterior vitreous detachment

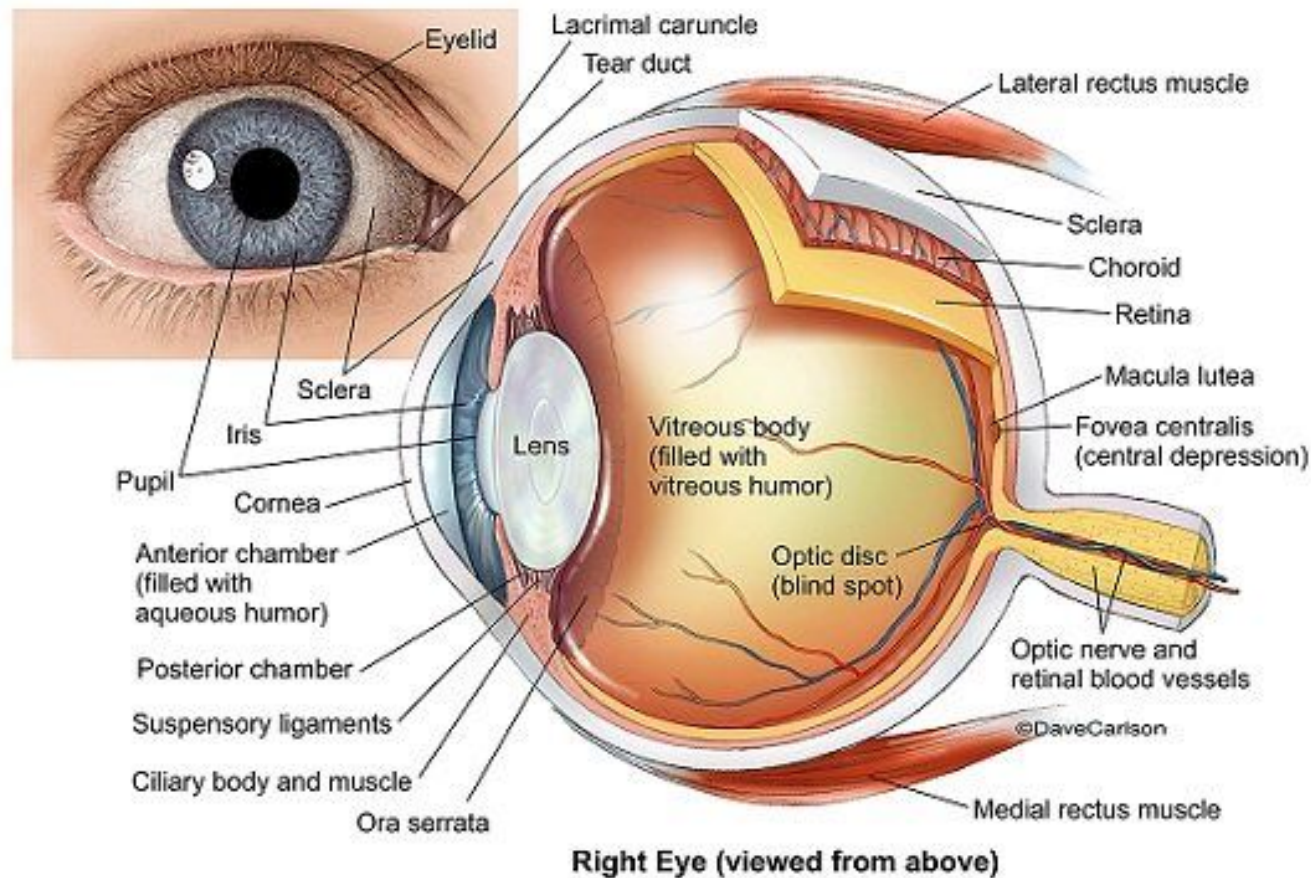


<https://www.allaboutvision.com/conditions/spotsfloaters.htm>

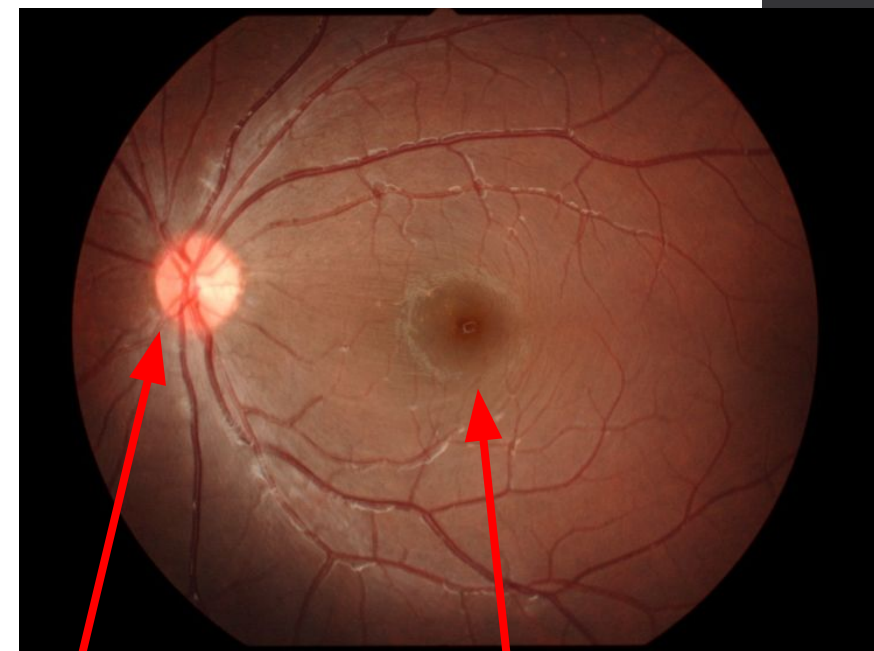
Retina - จอตา

- Thin and transparent structure
- 10 layers = RPE (retinal pigment epithelium) + neurosensory retina





Specialty Eye Care What is a Retina Specialist? - Specialty Eye Care (specialtyicare.com)



**Optic disc
เข้าประสาท**

**Macular &
fovea
จกรับภาพชัด**

Retina จอตา

- Convert the light waves into useful information including color, shape and motion.
- 10 layers = RPE + neurosensory retina

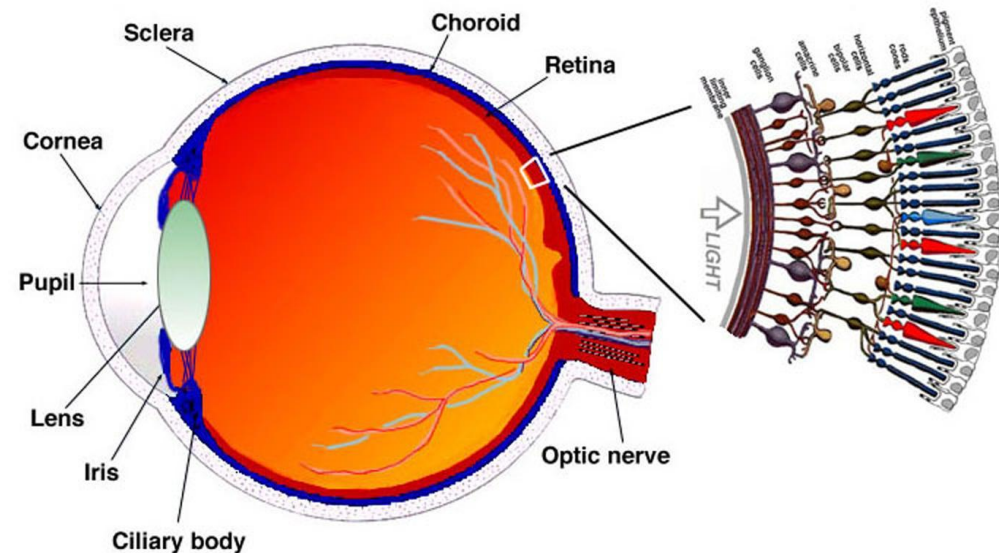
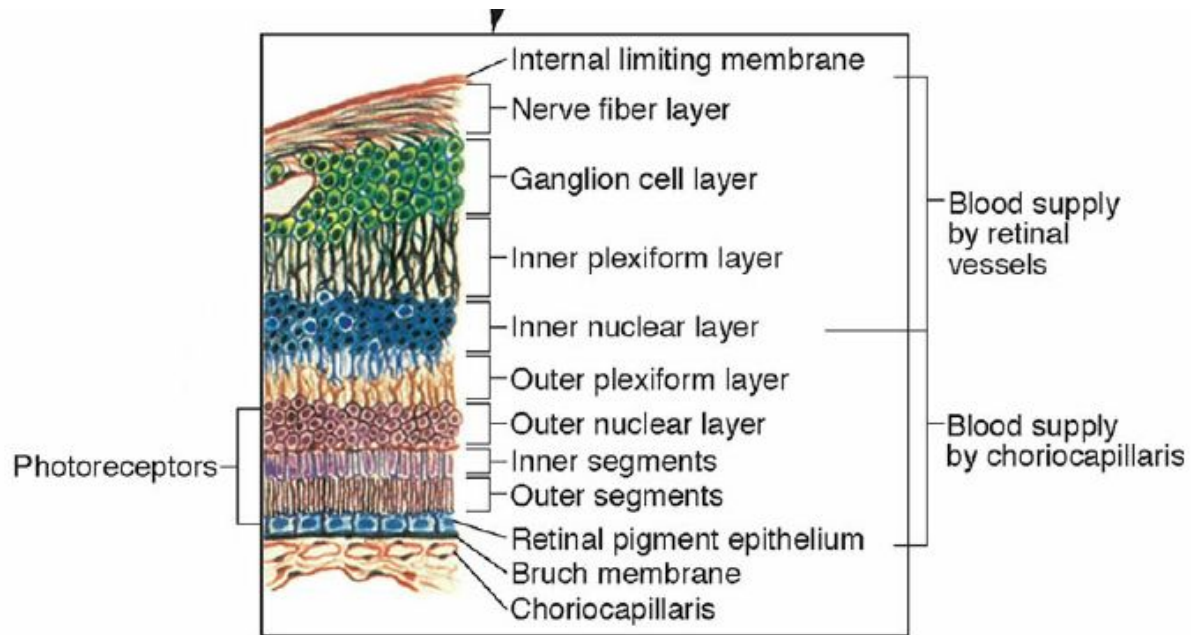
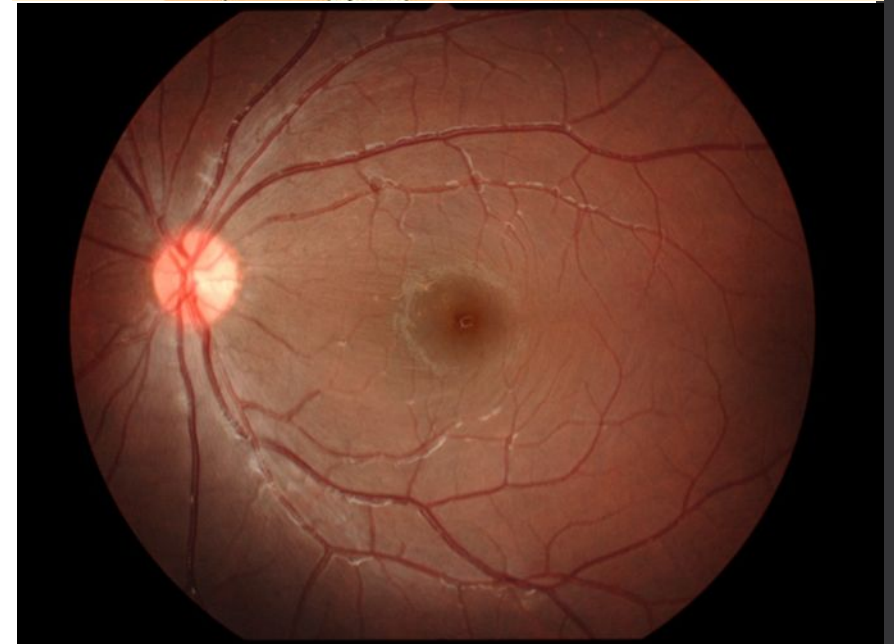
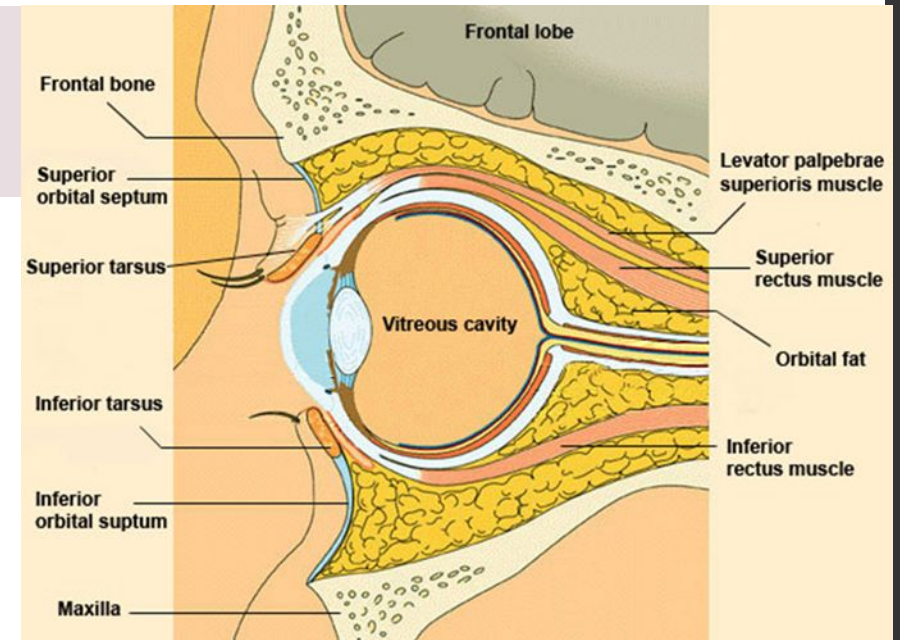


Fig. 1.1. A drawing of a section through the human eye with a schematic enlargement of the retina.

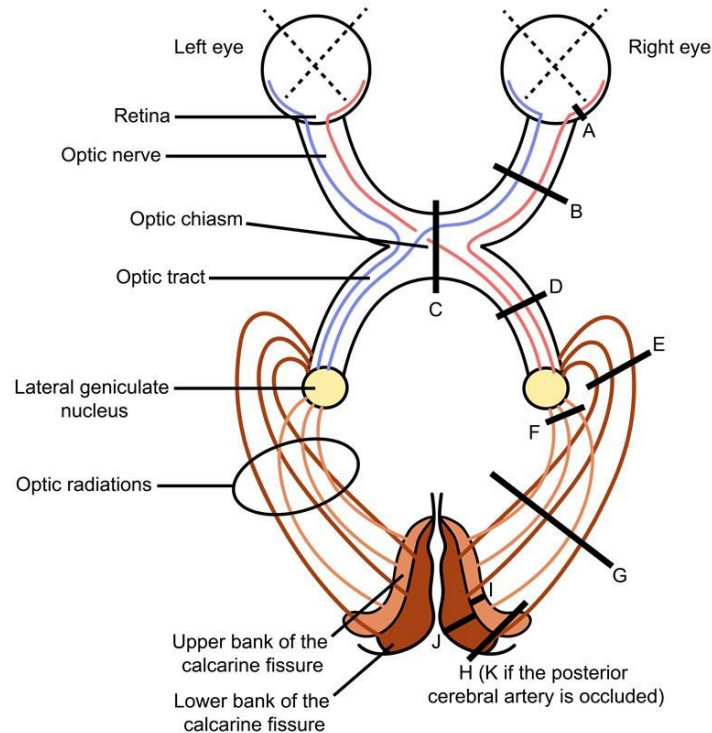
Optic nerve - เส้นประสาทตา

- Axon of ganglion cells
- Optic disc and physiologic cup
- Central retinal artery and vein
- 4 parts
 - intraocular part : optic disc, optic nerve head
 - intraorbital part
 - intracanalicular part
 - intracranial part



Visual pathway and visual field defect

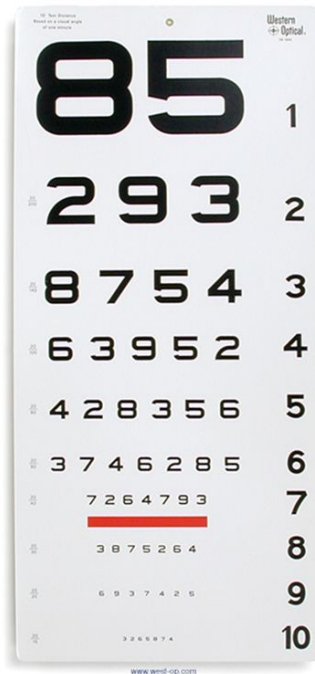
Visual Field Defects



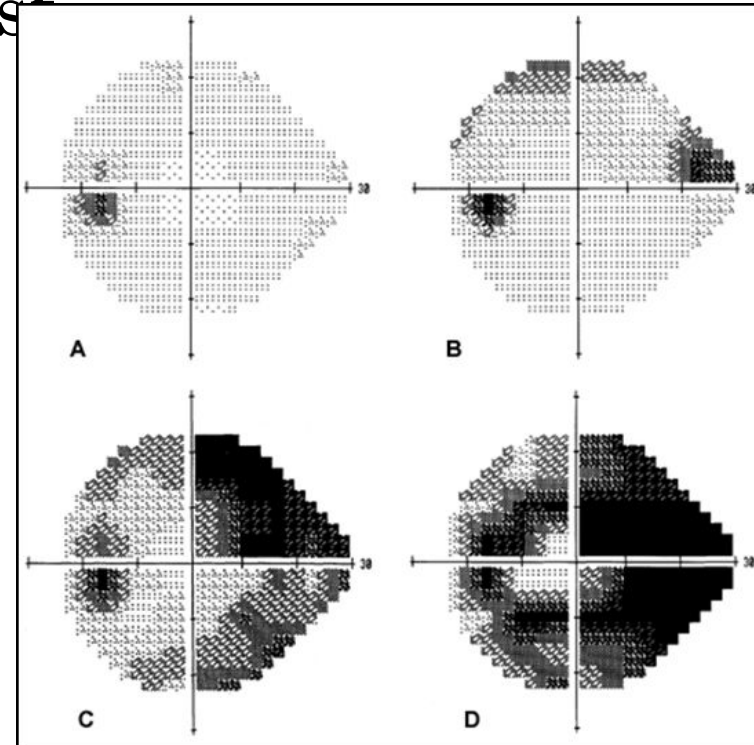
	Visual field deficit	
A) Central scotoma		
B) Monocular vision loss		
C) Bitemporal hemianopia		
D, G, & H) Contralateral homonymous hemianopia		
E & J) Contralateral superior quadrantanopia		
F & I) Contralateral inferior quadrantanopia		
K) Contralateral homonymous hemianopia with macular sparing		

การมองเห็น

- Visual acuity (VA) การมองเห็นตรงกลาง สิ่งที่เรากำลังมองอยู่
- การวัดสายตา: Snellen chart, ETDRS chart
 - VA 6/6, 6/6
 - ตัวเลขแรก-ระยะที่คนไข้
 - ตัวเลขที่สอง-ระยะที่คน



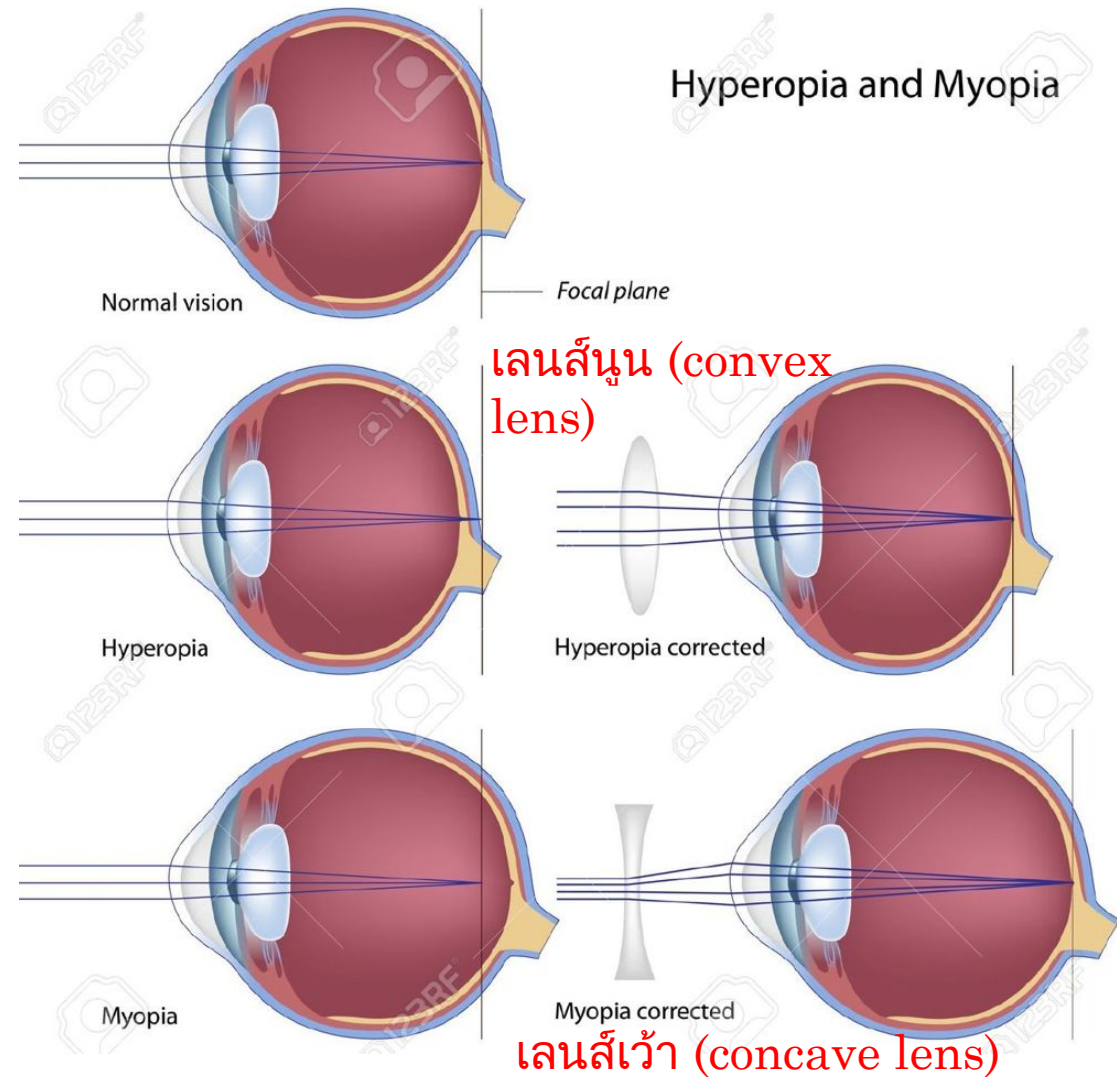
- Visual field ลานสายตา
- การวัดลานสายตา: confrontation test



Refractive system

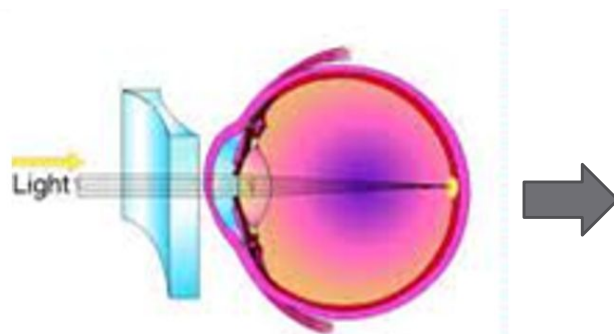
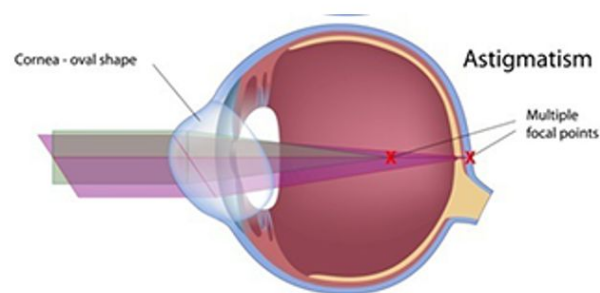
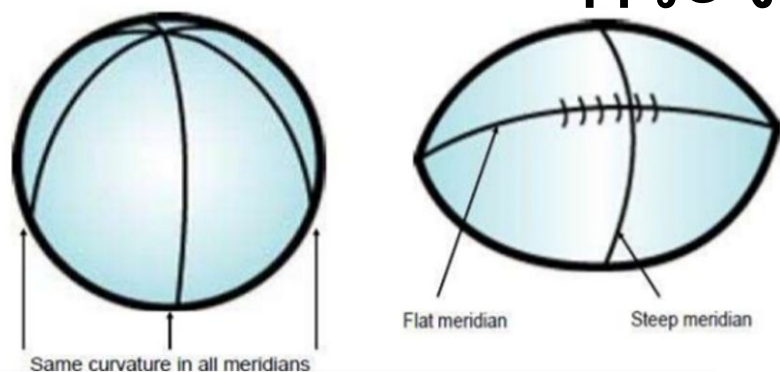
Refractive errors

- Hyperopia สายตายาว
- Myopia สายตาสั้น
- Astigmatism สายตาเอียง
- Presbyopia สายตายาวในคนอายุ 40 ปีขึ้นไป



Astigmatism สายตาเอียง

“ความโค้งของ
กระจกตาผิดปกติ”

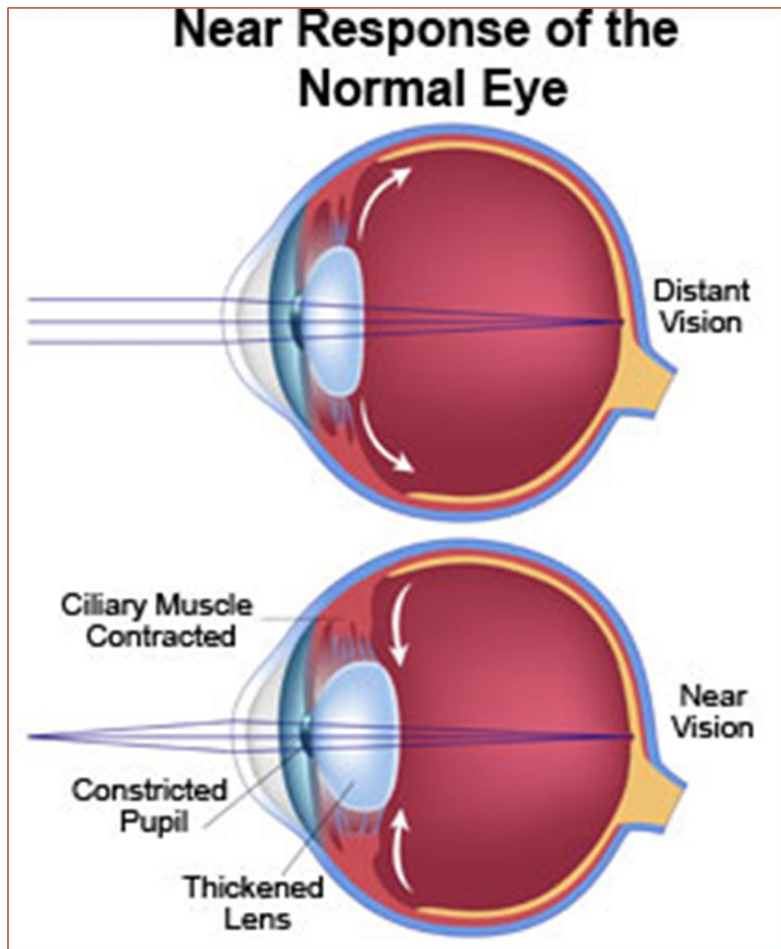


แก้ไขโดยใช้แว่นเลนส์
ทรงกระบอก
(cylindrical lens)

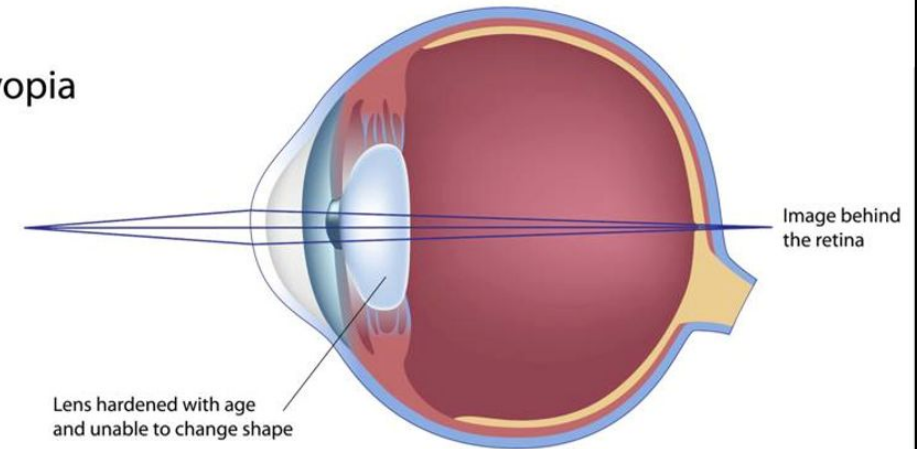
Presbyopia สายตายาวในคนอายุ 40ปีขึ้นไป

Focusing

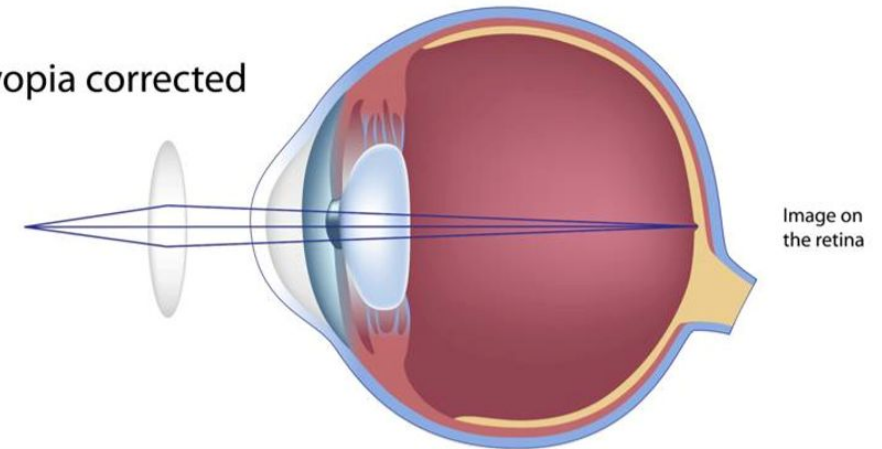
Accommodation
reflex



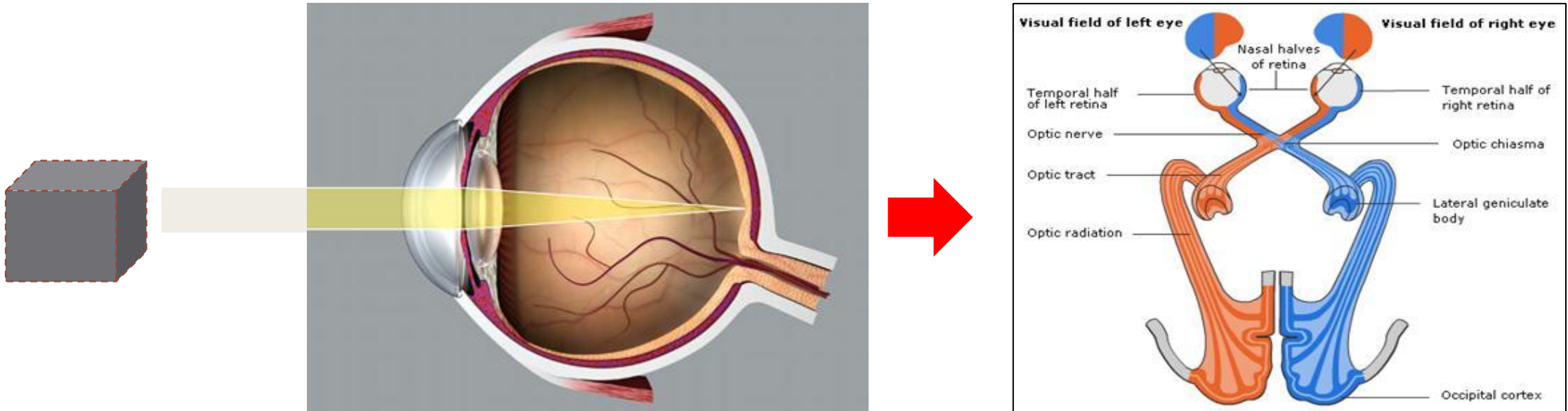
Presbyopia



Presbyopia corrected



การมองเห็น



1. แสงจากวัตถุ → **Ocular media** (cornea → aqueous → lens → vitreous)
2. หักเหและตกลงที่ retina → **Refractive system**
3. ภาพถูกส่งจาก retina สู่สมอง → **Visual pathway**

Bonus!

"Ocular Medication"

การใช้ยาทางจักษุ

Topical medications

- Eye drop
 - Solution
 - Suspension
- Eye ointment
- Eye gel

ใช้ง่าย
ออกฤทธิ์เฉพาะที่
ดูดซึมเข้าสู่ร่างกายน้อย

Local injection

- Subcutaneous injection
- Periocular injection
- Intraocular injection

Topical medications

Eye drop



Solution



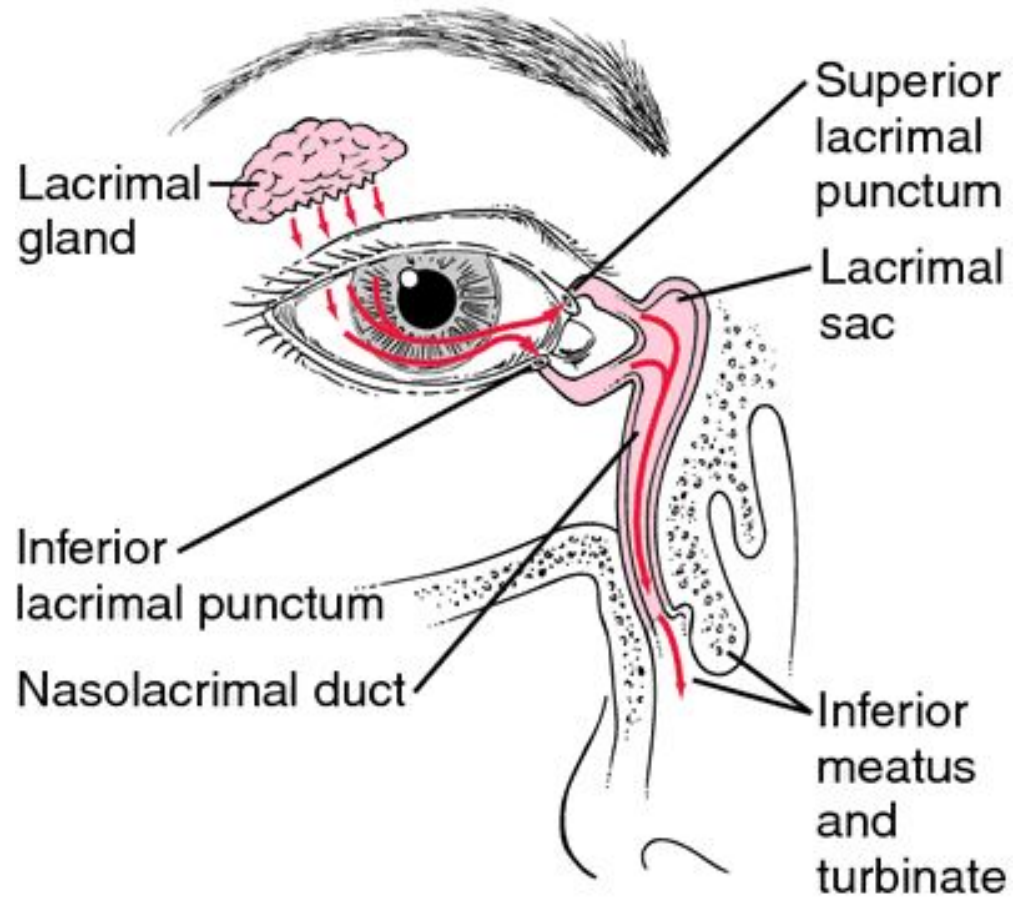
Suspension

Eye ointment



<http://www.eugeneeyecare.com/conditions-eye-drops/>

Lacrimal drainage system



Eye drop

- A standard eyedropper dispenses 0.05 ml (50 μ l) per drop
- Cul de sac volume = 30 μ L

Only one drop (50 μ l) is enough

Nasolacrimal duct compression



- Decrease systemic absorption
- Increase contact time

Eye drop & eye ointment administration



<https://www.glaucoma.org/treatment/eyedrop-tips.php>



<http://one.u-s-s.ru/eye-ointment-application/>

Local injection

- Subcutaneous injection
- Periocular injection
- Intraocular injection

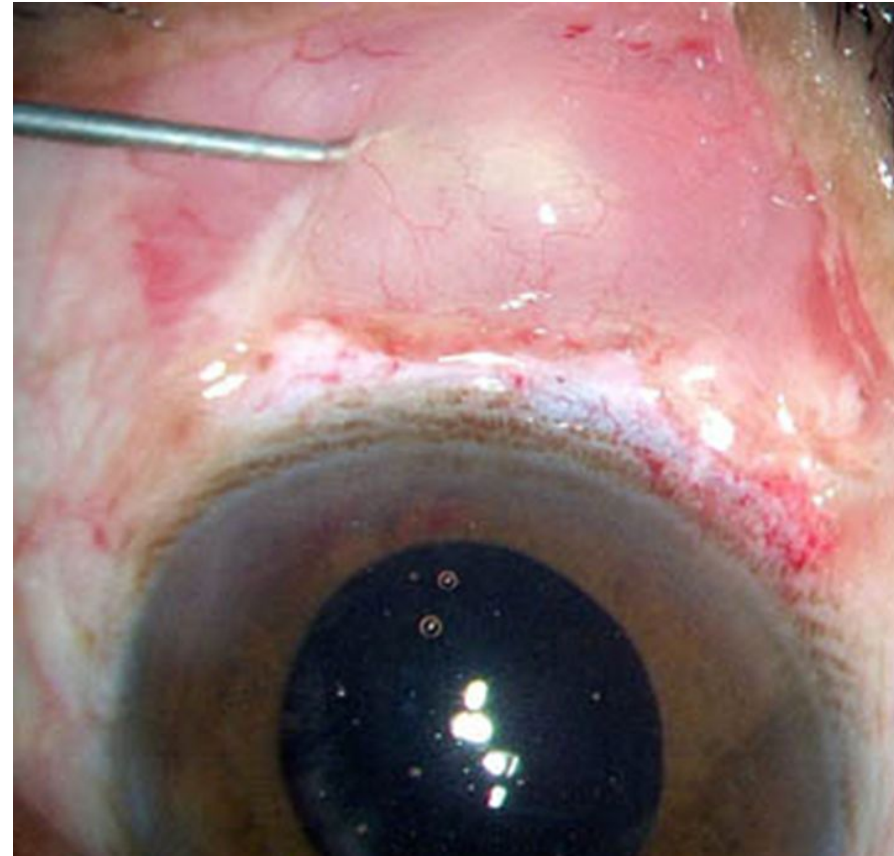
Subcutaneous injection



Periocular injection

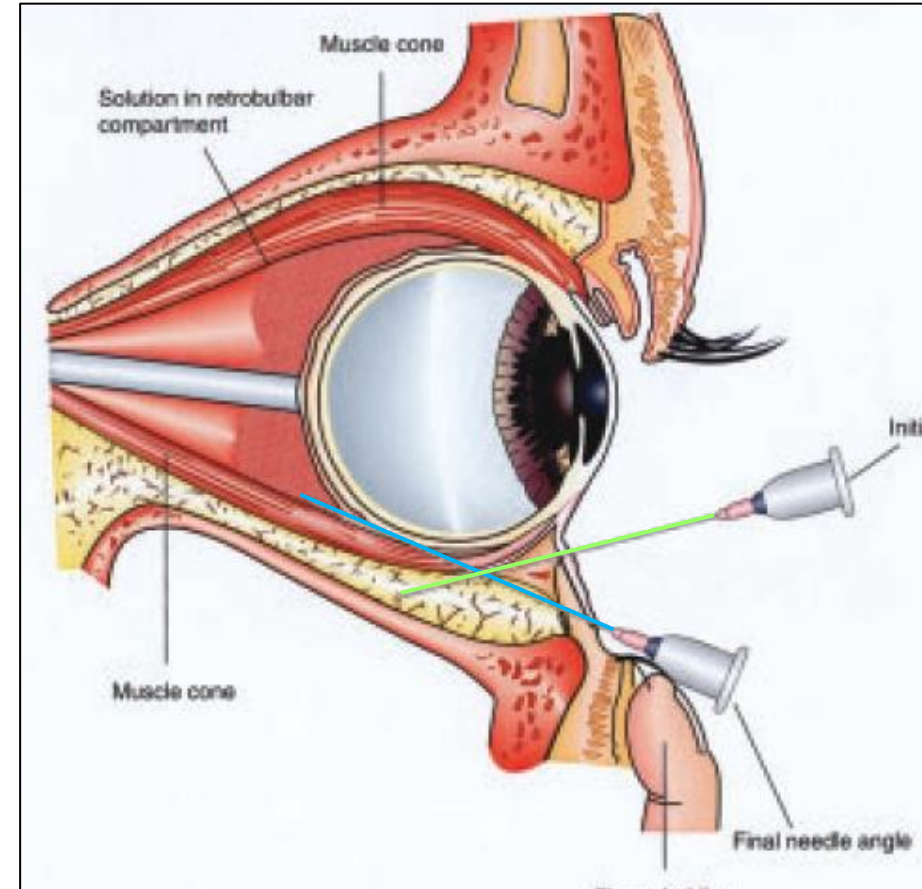
Subconjunctival injection

Subtenon injection



Periocular injection

Retrobulbar / Peribulbar injection

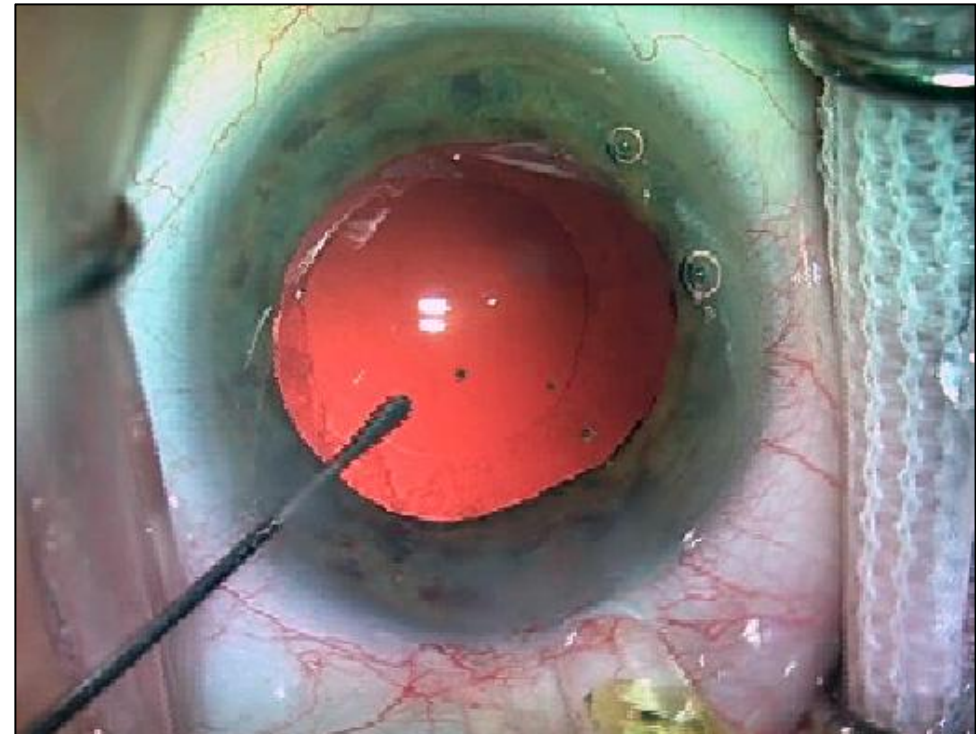


Intraocular injection

Intracameral injection



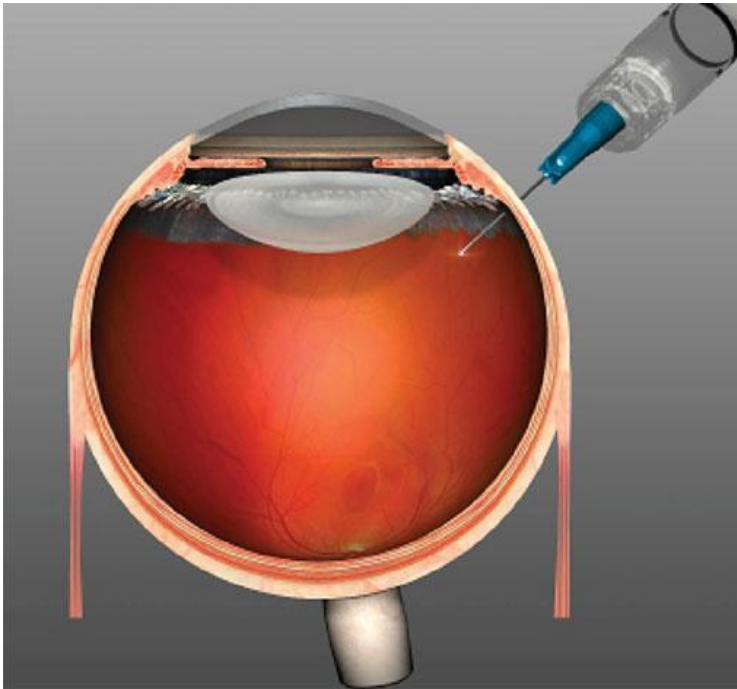
<https://www.opthalmologyweb.com>



<https://eyeworld-websupport911.netdna-ssl.com>

Intraocular injection

Intravitreal injection : pars plana injection



<https://eyeworld-websupport911.netdna-ssl.com>



<https://www.researchgate.net/>

THANK YOU

Q & A